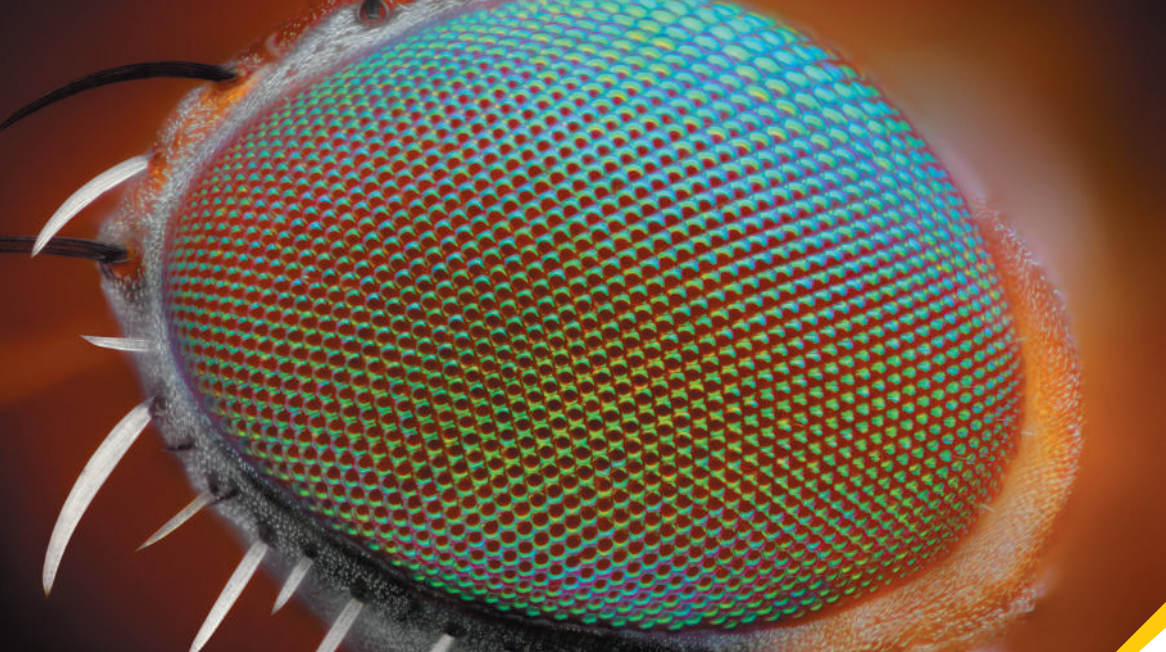




? Flies have *compound* eyes with thousands of miniature lenses. The overall diameter of the eye is about 1 mm, but each lens is only about $20\ \mu\text{m}$ in diameter and produces an individual image of a small region in the fly's field of view. Compared to the resolving power of the human eye (in which the light-gathering region is about 16 mm across), the ability of a fly's eye to resolve small details is (i) worse because the lenses are so small; (ii) worse because the eye as a whole is so small; (iii) better because the lenses are so small; (iv) better because the eye as a whole is so small; (v) about the same.

36 Diffraction

Everyone is used to the idea that sound bends around corners. If sound didn't behave this way, you couldn't hear a police siren that's out of sight or the speech of a person whose back is turned to you. But *light* can bend around corners as well. When light from a point source falls on a straightedge and casts a shadow, the edge of the shadow is never perfectly sharp. Some light appears in the area that we expect to be in the shadow, and we find alternating bright and dark fringes in the illuminated area. In general, light emerging from apertures doesn't precisely follow the predictions of the straight-line ray model of geometric optics.

The reason for these effects is that light, like sound, has wave characteristics. In Chapter 35 we studied the interference patterns that can arise when two light waves are combined. In this chapter we'll investigate interference effects due to combining *many* light waves. Such effects are referred to as *diffraction*. The behavior of waves after they pass through an aperture is an example of diffraction; each infinitesimal part of the aperture acts as a source of waves, and these waves interfere, producing a pattern of bright and dark fringes.

Similar patterns appear when light emerges from *arrays* of apertures. The nature of these patterns depends on the color of the light and the size and spacing of the apertures. Examples of this effect include the colors of iridescent butterflies and the "rainbow" you see reflected from the surface of a compact disc. We'll explore similar effects with x rays that are used to study the atomic structure of solids and liquids. Finally, we'll look at the physics of a *hologram*, a special kind of interference pattern used to form three-dimensional images.

36.1 FRESNEL AND FRAUNHOFER DIFFRACTION

According to geometric optics, when an opaque object is placed between a point light source and a screen, as in (Fig. 36.1, next page) the shadow of the object forms a perfectly sharp line. No light at all strikes the screen at points within the shadow, and the area outside the shadow is illuminated nearly uniformly. But as we saw in Chapter 35, the *wave* nature of light causes effects that can't be understood with geometric optics. An important class of such effects occurs when light strikes a barrier that has an aperture or an edge. The interference patterns formed in such a situation are grouped under the heading **diffraction**.

LEARNING OUTCOMES

In this chapter, you'll learn...

- 36.1 What happens when coherent light shines on an object with an edge or aperture.
- 36.2 How to understand the diffraction pattern formed when coherent light passes through a narrow slit.
- 36.3 How to calculate the intensity at various points in a single-slit diffraction pattern.
- 36.4 What happens when coherent light shines on an array of narrow, closely spaced slits.
- 36.5 How scientists use diffraction gratings for precise measurements of wavelength.
- 36.6 How x-ray diffraction reveals the arrangement of atoms in a crystal.
- 36.7 How diffraction sets limits on the smallest details that can be seen with an optical system.
- 36.8 How holograms work.

You'll need to review...

- 33.4, 33.7 Prisms and dispersion; Huygens's principle.
- 34.4, 34.5 Image formation by a lens; f -number.
- 35.1–35.3 Coherent light, two-slit interference, and phasors.

Figure 36.1 A point source of light illuminates a straightedge.

Geometric optics predicts that this situation should produce a sharp boundary between illumination and solid shadow.

That's NOT what really happens!

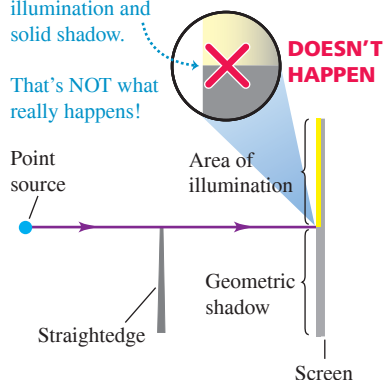


Figure 36.2 shows an example of diffraction. The photograph in Fig. 36.2a was made by placing a razor blade halfway between a pinhole, illuminated by monochromatic light, and a photographic film. The film recorded the shadow cast by the blade. Figure 36.2b is an enlargement of a region near the shadow of the right edge of the blade. The position of the *geometric* shadow line is indicated by arrows. The area outside the geometric shadow is bordered by alternating bright and dark bands. There is some light in the shadow region, although this is not very visible in the photograph. The first bright band in Fig. 36.2b, just to the right of the geometric shadow, is considerably brighter than in the region of uniform illumination to the extreme right. This simple experiment gives us some idea of the richness and complexity of diffraction.

We don't often observe diffraction patterns such as Fig. 36.2 in everyday life because most ordinary light sources are neither monochromatic nor point sources. If we use a white frosted light bulb instead of a point source to illuminate the razor blade in Fig. 36.2, each wavelength of the light from every point of the bulb forms its own diffraction pattern, but the patterns overlap so much that we can't see any individual pattern.

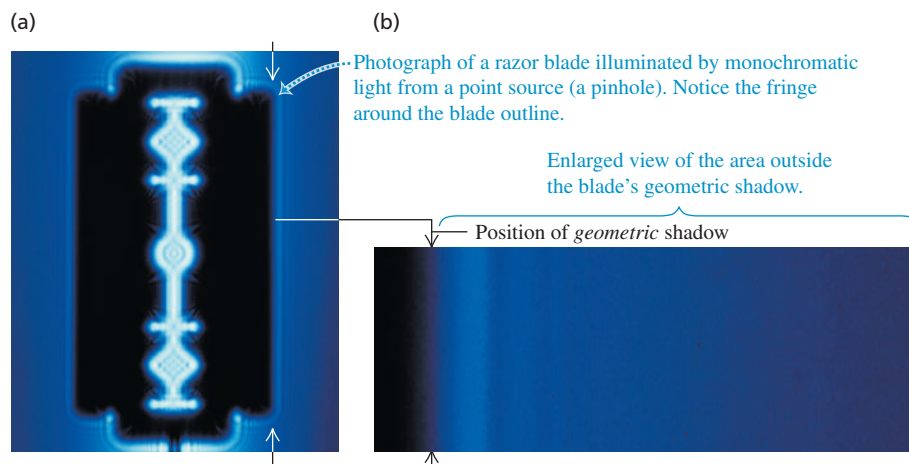
Diffraction and Huygens's Principle

We can use Huygens's principle (see Section 33.7) to analyze diffraction patterns. This principle states that we can consider every point of a wave front as a source of secondary wavelets. These spread out in all directions with a speed equal to the speed of propagation of the wave. The position of the wave front at any later time is the *envelope* of the secondary wavelets at that time. To find the resultant displacement at any point, we use the superposition principle to combine all the individual displacements produced by these secondary waves.

In Fig. 36.1, both the point source and the screen are relatively close to the obstacle forming the diffraction pattern. This situation is described as *near-field diffraction* or **Fresnel diffraction**, pronounced “Freh-nell” (after the French scientist Augustin Jean Fresnel, 1788–1827). By contrast, we use the term **Fraunhofer diffraction** (after the German physicist Joseph von Fraunhofer, 1787–1826) for situations in which the source, obstacle, and screen are far enough apart that we can consider all lines from the source to the obstacle to be parallel, and can likewise consider all lines from the obstacle to a given point on the screen to be parallel. We'll restrict the following discussion to Fraunhofer diffraction, which is usually simpler to analyze in detail than Fresnel diffraction.

Diffraction is sometimes described as “the bending of light around an obstacle.” But the process that causes diffraction is present in the propagation of *every* wave. When part of the wave is cut off by some obstacle, we observe diffraction effects that result from interference of the remaining parts of the wave fronts. Optical instruments typically use only a limited portion of a wave; for example, a telescope uses only the part of a wave that is admitted by its objective lens or mirror. Thus diffraction plays a role in nearly all optical phenomena.

Figure 36.2 An example of diffraction.



Finally, we emphasize that there is no fundamental distinction between *interference* and *diffraction*. In Chapter 35 we used the term *interference* for effects involving waves from a small number of sources, usually two. *Diffraction* usually involves a continuous distribution of Huygens's wavelets across the area of an aperture, or a very large number of sources or apertures. But both interference and diffraction are consequences of superposition and Huygens's principle.

TEST YOUR UNDERSTANDING OF SECTION 36.1 Can sound waves undergo diffraction around an edge?

ANSWER

I yes When you hear the voice of someone standing around a corner, you are hearing sound waves that underwent diffraction. If there were no diffraction or reflection of sound, you could hear sounds only from objects that were in plain view.

36.2 DIFFRACTION FROM A SINGLE SLIT

In this section we'll discuss the diffraction pattern formed by plane-wave (parallel-ray) monochromatic light when it emerges from a long, narrow slit, as shown in **Fig. 36.3**. We call the narrow dimension the *width*, even though in this figure it is a vertical dimension.

According to geometric optics, the transmitted beam should have the same cross section as the slit, as in Fig. 36.3a. What is *actually* observed is the pattern shown in Fig. 36.3b. The beam spreads out vertically after passing through the slit. The diffraction pattern consists of a central bright band, which may be much broader than the width of the slit, bordered by alternating dark and bright bands with rapidly decreasing intensity. About 85% of the power in the transmitted beam is in the central bright band, whose width is *inversely* proportional to the slit width. **In general, the narrower the slit, the broader the entire diffraction pattern.** (The *horizontal* spreading of the beam in Fig. 36.3b is negligible because the horizontal dimension of the slit is relatively large.) You can observe a similar diffraction pattern by looking at a point source, such as a distant street light, through a narrow slit formed between your two thumbs held in front of your eye; the retina of your eye acts as the screen.

Single-Slit Diffraction: Locating the Dark Fringes

Figure 36.4 (next page) shows a side view of the same setup; the long sides of the slit are perpendicular to the figure, and plane waves are incident on the slit from the left. According to Huygens's principle, each element of area of the slit opening can be considered as a source of secondary waves. In particular, imagine dividing the slit into several narrow strips of equal width, parallel to the long edges and perpendicular to the page. Figure 36.4a shows two such strips. Cylindrical secondary wavelets, shown in cross section, spread out from each strip.

Figure 36.3 (a) The “shadow” of a horizontal slit as incorrectly predicted by geometric optics.

(b) A horizontal slit actually produces a diffraction pattern. The slit width has been greatly exaggerated.

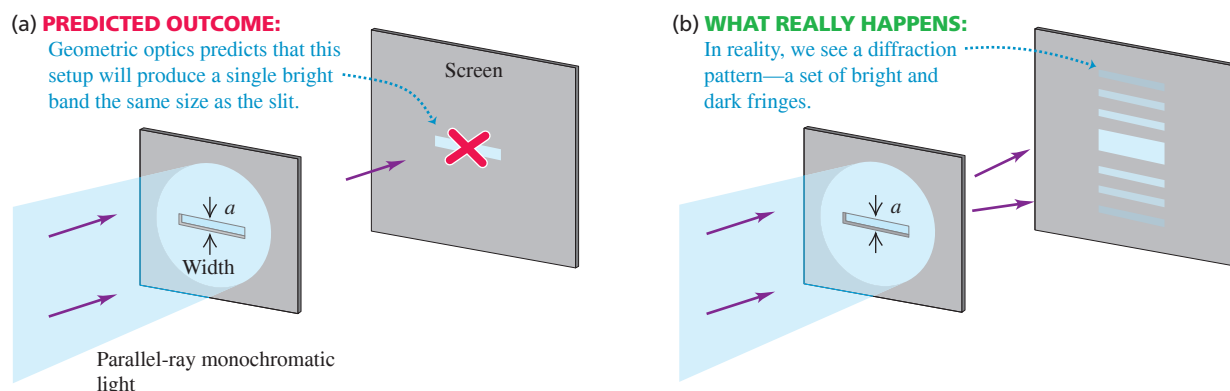
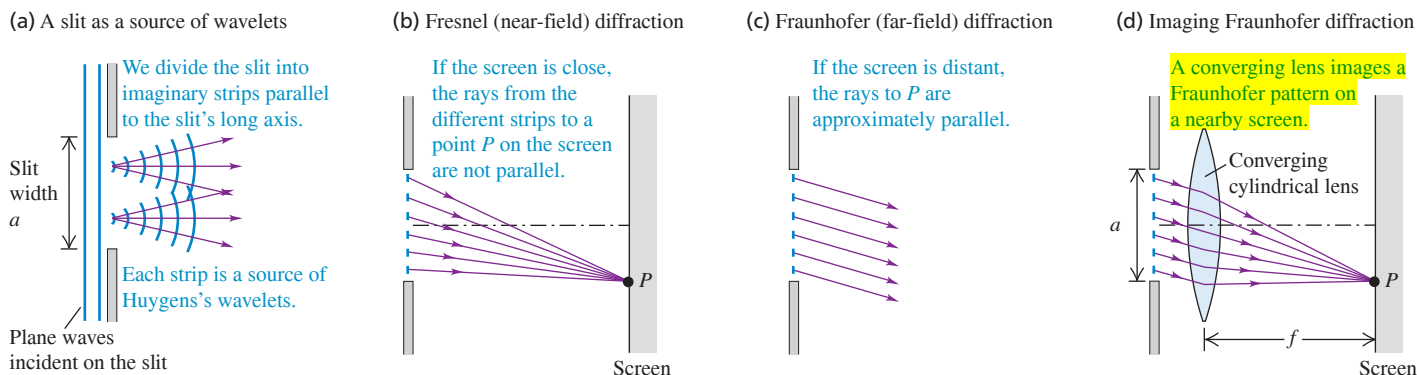


Figure 36.4 Diffraction by a single rectangular slit. The long sides of the slit are perpendicular to the figure.

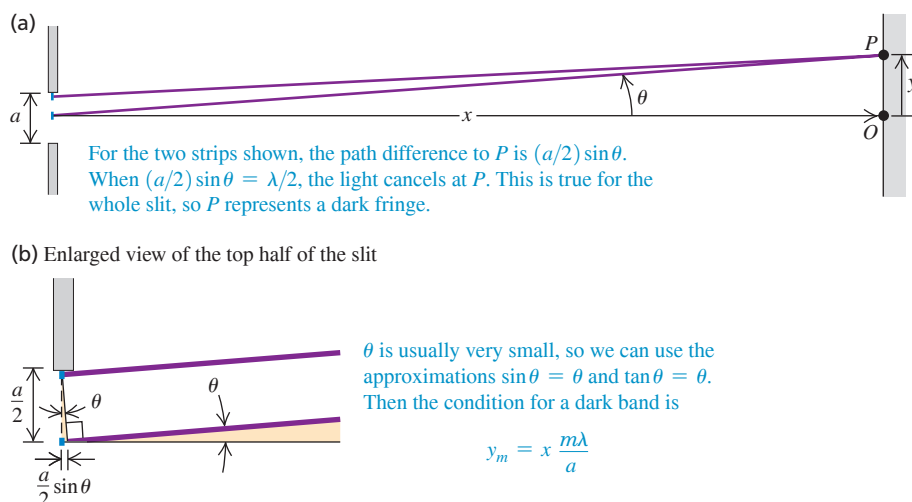


In Fig. 36.4b a screen is placed to the right of the slit. We can calculate the resultant intensity at a point P on the screen by adding the contributions from the individual wavelets, taking proper account of their various phases and amplitudes. It's easiest to do this calculation if we assume that the screen is far enough away that all the rays from various parts of the slit to a particular point P on the screen are parallel, as in Fig. 36.4c. An equivalent situation is Fig. 36.4d, in which the rays to the lens are parallel and the lens forms a reduced image of the same pattern that would be formed on an infinitely distant screen without the lens. We might expect that the various light paths through the lens would introduce additional phase shifts, but in fact it can be shown that all the paths have *equal* phase shifts, so this is not a problem.

The situation of Fig. 36.4b is Fresnel diffraction; those in Figs. 36.4c and 36.4d, where the outgoing rays are considered parallel, are Fraunhofer diffraction. We can derive quite simply the most important characteristics of the Fraunhofer diffraction pattern from a single slit. First consider two narrow strips, one just below the top edge of the drawing of the slit and one at its center, shown in end view in **Fig. 36.5**. The difference in path length to point P is $(a/2)\sin\theta$, where a is the slit width and θ is the angle between the perpendicular to the slit and a line from the center of the slit to P . Suppose this path difference happens to be equal to $\lambda/2$; then light from these two strips arrives at point P with a half-cycle phase difference, and cancellation occurs.

Similarly, light from two strips immediately *below* the two in the figure also arrives at P a half-cycle out of phase. In fact, the light from *every* strip in the top half of the slit cancels out the light from a corresponding strip in the bottom half. Hence the combined

Figure 36.5 Side view of a horizontal slit. When the distance x to the screen is much greater than the slit width a , the rays from a distance $a/2$ apart may be considered parallel.



light from the entire slit completely cancels at P , giving a dark fringe in the interference pattern. A dark fringe occurs whenever

$$\frac{a}{2} \sin \theta = \pm \frac{\lambda}{2} \quad \text{or} \quad \sin \theta = \pm \frac{\lambda}{a} \quad (36.1)$$

The plus-or-minus ($\pm$) sign in Eq. (36.1) says that there are symmetric dark fringes above and below point O in Fig. 36.5a. The upper fringe ($\theta > 0$) occurs at a point P where light from the bottom half of the slit travels $\lambda/2$ farther to P than does light from the top half; the lower fringe ($\theta < 0$) occurs where light from the *top* half travels $\lambda/2$ farther than light from the *bottom* half.

We may also divide the slit into quarters, sixths, and so on, and use the above argument to show that a dark fringe occurs whenever $\sin \theta = \pm 2\lambda/a$, $\pm 3\lambda/a$, and so on. Thus the condition for a *dark* fringe is

Dark fringes, single-slit diffraction:

Angle of line from center of slit to m th dark fringe on screen

$$\sin \theta = \pm \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \dots) \quad (36.2)$$

Slit width Wavelength

For example, if the slit width is equal to ten wavelengths ($a = 10\lambda$), dark fringes occur at $\sin \theta = \pm \frac{1}{10}$, $\pm \frac{2}{10}$, $\pm \frac{3}{10}$, $\dots$. Between the dark fringes are bright fringes. Note that $\sin \theta = 0$ corresponds to a *bright* band; in this case, light from the entire slit arrives at P in phase. Thus it would be wrong to put $m = 0$ in Eq. (36.2).

With light, the wavelength λ is of the order of $500 \text{ nm} = 5 \times 10^{-7} \text{ m}$. This is often much smaller than the slit width a ; a typical slit width is $10^{-2} \text{ cm} = 10^{-4} \text{ m}$. Therefore the values of θ in Eq. (36.2) are often so small that the approximation $\sin \theta \approx \theta$ (where θ is in radians) is a very good one. In that case we can rewrite this equation as

$$\theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \dots) \quad (\text{for small angles } \theta \text{ in radians})$$

Also, if the distance from slit to screen is x , as in Fig. 36.5a, and the vertical distance of the m th dark band from the center of the pattern is y_m , then $\tan \theta = y_m/x$. For small θ we may also approximate $\tan \theta$ by θ (in radians). We then find

$$y_m = x \frac{m\lambda}{a} \quad (\text{for } y_m \ll x) \quad (36.3)$$

Figure 36.6 is a photograph of a single-slit diffraction pattern with the $m = \pm 1$, ± 2 , and ± 3 minima labeled. The central bright fringe is wider than the other bright fringes; in the small-angle approximation used in Eq. (36.3), it is exactly twice as wide.

CAUTION Single-slit diffraction vs. two-slit interference Equation (36.3) has the same form as the equation for the two-slit pattern, Eq. (35.6), except that in Eq. (36.3) we use x rather than R for the distance to the screen. But Eq. (36.3) gives the positions of the *dark* fringes in a *single-slit* pattern rather than the *bright* fringes in a *double-slit* pattern. Also, $m = 0$ in Eq. (36.2) is *not* a dark fringe. Finally, note that Eq. (36.2) gives the angle θ from the m th dark fringe to the center of the diffraction pattern—not the angle from a dark fringe on one side of the pattern to the corresponding dark fringe on the other side. Be careful! ■

Figure 36.6 Photograph of the Fraunhofer diffraction pattern of a single horizontal slit.



EXAMPLE 36.1 Single-slit diffraction

WITH VARIATION PROBLEMS

You pass 633 nm laser light through a narrow slit and observe the diffraction pattern on a screen 6.0 m away. The distance on the screen between the centers of the first minima on either side of the central bright fringe is 32 mm (**Fig. 36.7**, next page). How wide is the slit?

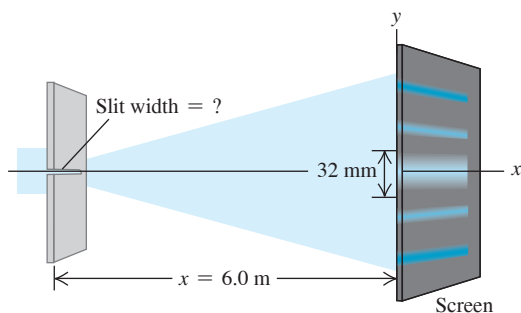
IDENTIFY and SET UP This problem involves the relationship between the positions of dark fringes in a single-slit diffraction pattern and the slit width a (our target variable). The distances between fringes on the screen are much smaller than the slit-to-screen distance, so the angle θ shown in Fig. 36.5a is very small and we can use Eq. (36.3) to solve for a .

EXECUTE The first minimum corresponds to $m = 1$ in Eq. (36.3). The distance y_1 from the central maximum to the first minimum on either side is half the distance between the two first minima, so $y_1 = (32 \text{ mm})/2 = 16 \text{ mm}$. Solving Eq. (36.3) for a , we find

$$a = \frac{x\lambda}{y_1} = \frac{(6.0 \text{ m})(633 \times 10^{-9} \text{ m})}{16 \times 10^{-3} \text{ m}} = 2.4 \times 10^{-4} \text{ m} = 0.24 \text{ mm}$$

Continued

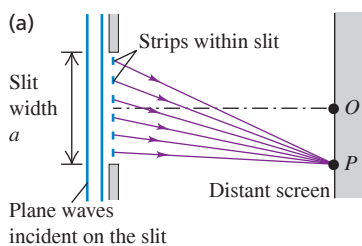
Figure 36.7 A single-slit diffraction experiment.



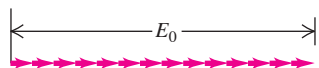
EVALUATE The angle θ is small only if the wavelength is small compared to the slit width. Since $\lambda = 633 \text{ nm} = 6.33 \times 10^{-7} \text{ m}$ and we have found $a = 0.24 \text{ mm} = 2.4 \times 10^{-4} \text{ m}$, our result is consistent with this: The wavelength is $(6.33 \times 10^{-7} \text{ m}) / (2.4 \times 10^{-4} \text{ m}) = 0.0026$ as large as the slit width. Can you show that the distance between the *second* minima on either side is $2(32 \text{ mm}) = 64 \text{ mm}$, and so on?

KEYCONCEPT In single-slit diffraction, wavelets that emerge from different parts of the slit interfere and produce a pattern of bright and dark fringes. Equations (36.2) and (36.3) give the positions of the dark fringes where destructive interference occurs.

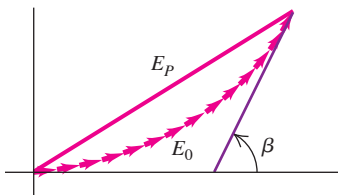
Figure 36.8 Using phasor diagrams to find the amplitude of the $\vec{E}$ field in single-slit diffraction. Each phasor represents the $\vec{E}$ field from a single strip within the slit.



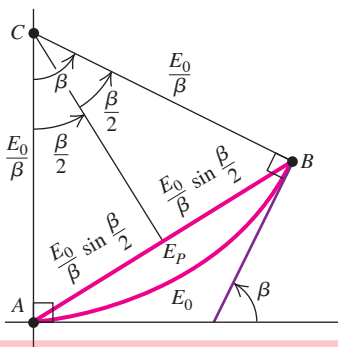
(b) At the center of the diffraction pattern (point O), the phasors from all strips within the slit are in phase.



(c) Phasor diagram at a point slightly off the center of the pattern; β = total phase difference between the first and last phasors.



(d) As in (c), but in the limit that the slit is subdivided into infinitely many strips



TEST YOUR UNDERSTANDING OF SECTION 36.2 Rank the following single-slit diffraction experiments in order of the size of the angle from the center of the diffraction pattern to the first dark fringe, from largest to smallest: (i) Wavelength 400 nm, slit width 0.20 mm; (ii) wavelength 600 nm, slit width 0.20 mm; (iii) wavelength 400 nm, slit width 0.30 mm; (iv) wavelength 600 nm, slit width 0.30 mm.

ANSWER

(i) $\lambda = 400 \text{ nm}$, $a = 0.20 \text{ mm}$; (ii) $\lambda = 600 \text{ nm}$, $a = 0.20 \text{ mm}$; (iii) $\lambda = 400 \text{ nm}$, $a = 0.30 \text{ mm}$; (iv) $\lambda = 600 \text{ nm}$, $a = 0.30 \text{ mm}$. The ratio λ/a in each case is the value of θ . The larger the value of the ratio λ/a , the larger the value of $\sin \theta$ and hence $m = 1$, or $\sin \theta = \lambda/a$. The angle θ of the first dark fringe is given by Eq. (36.2) with

36.3 INTENSITY IN THE SINGLE-SLIT PATTERN

We can derive an expression for the intensity distribution for the single-slit diffraction pattern by the same phasor-addition method that we used in Section 35.3 for the two-slit interference pattern. We again imagine a plane wave front at the slit subdivided into a large number of strips. We superpose the contributions of the Huygens wavelets from all the strips at a point P on a distant screen at an angle θ from the normal to the slit plane (Fig. 36.8a). To do this, we use a phasor to represent the sinusoidally varying $\vec{E}$ field from each strip. The magnitude of the vector sum of the phasors at each point P is the amplitude E_P of the total $\vec{E}$ field at that point. The intensity at P is proportional to E_P^2 .

At the point O shown in Fig. 36.8a, corresponding to the center of the pattern where $\theta = 0$, there are negligible path differences for $x \gg a$; the phasors are all essentially in phase (that is, have the same direction). In Fig. 36.8b we draw the phasors at time $t = 0$ and denote the resultant amplitude at O by E_0 . In this illustration we have divided the slit into 14 strips.

Now consider wavelets arriving from different strips at point P in Fig. 36.8a, at an angle θ from point O . Because of the differences in path length, there are now phase differences between wavelets coming from adjacent strips; the corresponding phasor diagram is shown in Fig. 36.8c. The vector sum of the phasors is now part of the perimeter of a many-sided polygon, and E_P , the amplitude of the resultant electric field at P , is the chord. The angle β is the total phase difference between the wave received at P from the top strip of Fig. 36.8a and the wave received at P from the bottom strip.

We may imagine dividing the slit into narrower and narrower strips. In the limit that there is an infinite number of infinitesimally narrow strips, the curved trail of phasors becomes an arc of a circle (Fig. 36.8d), with arc length equal to the length E_0 in Fig. 36.8b. The center C of this arc is found by constructing perpendiculars at A and B . From the relationship among arc length, radius, and angle, the radius of the arc is E_0/β ; the amplitude

E_P of the resultant electric field at P is equal to the chord AB , which is $2(E_0/\beta) \sin(\beta/2)$. (Note that β *must* be in radians!) We then have

$$E_P = E_0 \frac{\sin(\beta/2)}{\beta/2} \quad (\text{amplitude in single-slit diffraction}) \quad (36.4)$$

The intensity at each point on the screen is proportional to the square of the amplitude given by Eq. (36.4). If I_0 is the intensity in the straight-ahead direction where $\theta = 0$ and $\beta = 0$, then the intensity I at any point is

$$I = I_0 \left[\frac{\sin(\beta/2)}{\beta/2} \right]^2 \quad (\text{intensity in single-slit diffraction}) \quad (36.5)$$

We can express the phase difference β in terms of geometric quantities, as we did for the two-slit pattern. From Eq. (35.11) the phase difference is $2\pi/\lambda$ times the path difference. Figure 36.5 shows that the path difference between the ray from the top of the slit and the ray from the middle of the slit is $(a/2) \sin \theta$. The path difference between the rays from the top of the slit and the bottom of the slit is twice this, so

$$\beta = \frac{2\pi}{\lambda} a \sin \theta \quad (36.6)$$

and Eq. (36.5) becomes

$$I = I_0 \left\{ \frac{\sin \left[\pi a (\sin \theta) / \lambda \right]}{\pi a (\sin \theta) / \lambda} \right\}^2 \quad (36.7)$$

Angle of line from center of slit to position on screen
Intensity in single-slit diffraction
Intensity at $\theta = 0$
Slit width
Wavelength

This equation expresses the intensity directly in terms of the angle θ . In many calculations it is easier first to calculate the phase angle β , from Eq. (36.6), and then to use Eq. (36.5).

Equation (36.7) is plotted in **Fig. 36.9a**. Note that the central intensity peak is much larger than any of the others. This means that most of the power in the wave remains within an angle θ from the perpendicular to the slit, where $\sin \theta = \lambda/a$ (the first diffraction minimum). You can see this easily in Fig. 36.9b, which is a photograph of water waves undergoing single-slit diffraction. Note also that the peak intensities in Fig. 36.9a decrease rapidly as we go away from the center of the pattern. (Compare Fig. 36.6, which shows a single-slit diffraction pattern for light.)

The dark fringes in the pattern are the places where $I = 0$. These occur at points for which the numerator of Eq. (36.5) is zero so that β is a multiple of 2π . From Eq. (36.6) this corresponds to

$$\begin{aligned} \frac{a \sin \theta}{\lambda} &= m \quad (m = \pm 1, \pm 2, \dots) \\ \sin \theta &= \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \dots) \end{aligned} \quad (36.8)$$

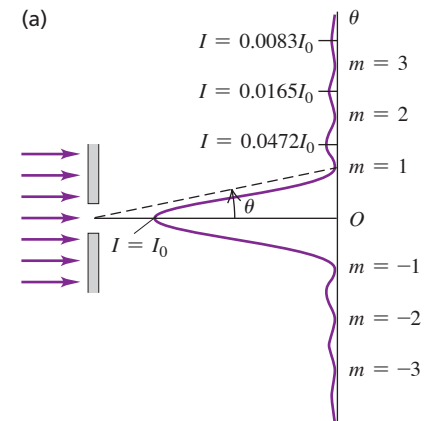
This agrees with our previous result, Eq. (36.2). Note again that $\beta = 0$ (corresponding to $\theta = 0$) is *not* a minimum. Equation (36.5) is indeterminate at $\beta = 0$, but we can evaluate the limit as $\beta \rightarrow 0$ by using L'Hôpital's rule. We find that at $\beta = 0$, $I = I_0$, as we should expect.

Intensity Maxima in the Single-Slit Pattern

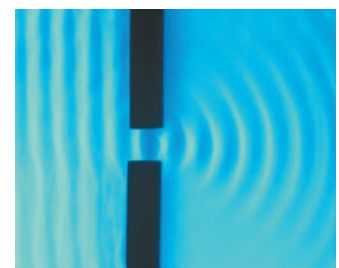
We can also use Eq. (36.5) to calculate the positions of the peaks, or *intensity maxima*, and the intensities at these peaks. This is not quite as simple as it may appear. We might expect the peaks to occur where the sine function reaches the value ± 1 —namely, where $\beta = \pm \pi, \pm 3\pi, \pm 5\pi$, or in general,

$$\beta \approx \pm (2m + 1)\pi \quad (m = 0, 1, 2, \dots) \quad (36.9)$$

Figure 36.9 (a) Intensity versus angle in single-slit diffraction. The values of m label intensity minima given by Eq. (36.8). Most of the wave power goes into the central intensity peak (between the $m = 1$ and $m = -1$ intensity minima). (b) These water waves passing through a small aperture behave exactly like light waves in single-slit diffraction. Only the diffracted waves within the central intensity peak are visible; the waves at larger angles are too faint to see.



(b)



This is *approximately* correct, but because of the factor $(\beta/2)^2$ in the denominator of Eq. (36.5), the maxima don't occur precisely at these points. When we take the derivative of Eq. (36.5) with respect to β and set it equal to zero to try to find the maxima and minima, we get a transcendental equation that has to be solved numerically. In fact there is *no* maximum near $\beta = \pm\pi$. The first maxima on either side of the central maximum, near $\beta = \pm 3\pi$, actually occur at $\pm 2.860\pi$. The second side maxima, near $\beta = \pm 5\pi$, are actually at $\pm 4.918\pi$, and so on. The error in Eq. (36.9) vanishes in the limit of large m —that is, for intensity maxima far from the center of the pattern.

To find the intensities at the side maxima, we substitute these values of β back into Eq. (36.5). Using the approximate expression in Eq. (36.9), we get

$$I_m \approx \frac{I_0}{\left(m + \frac{1}{2}\right)^2 \pi^2} \quad (36.10)$$

where I_m is the intensity of the m th side maximum and I_0 is the intensity of the central maximum. Equation (36.10) gives the series of intensities

$$0.0450I_0 \quad 0.0162I_0 \quad 0.0083I_0$$

and so on. As we have pointed out, this equation is only approximately correct. The actual intensities of the side maxima turn out to be

$$0.0472I_0 \quad 0.0165I_0 \quad 0.0083I_0 \quad \dots$$

These intensities decrease very rapidly, as Fig. 36.9a also shows. Even the first side maxima have less than 5% of the intensity of the central maximum.

Width of the Single-Slit Pattern

For small angles the angular spread of the diffraction pattern is inversely proportional to the ratio of the slit width a to the wavelength λ . **Figure 36.10** shows graphs of intensity I as a function of the angle θ for three values of the ratio a/λ .

With light waves, the wavelength λ is often much smaller than the slit width a , and the values of θ in Eqs. (36.6) and (36.7) are so small that the approximation $\sin\theta = \theta$ is very good. With this approximation the position θ_1 of the first ($m = 1$) minimum, corresponding to $\beta/2 = \pi$, is, from Eq. (36.7),

$$\theta_1 = \frac{\lambda}{a} \quad (36.11)$$

This characterizes the width (angular spread) of the central maximum, and we see that it is *inversely* proportional to the slit width a . When the small-angle approximation is valid, the central maximum is exactly twice as wide as each side maximum. When a is of the order of a centimeter or more, θ_1 is so small that we can consider practically all the light to be concentrated at the geometrical focus. But when a is less than λ , the central maximum spreads over 180° , and the fringe pattern is not seen at all.

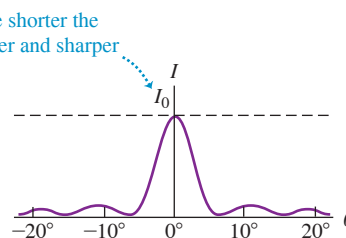
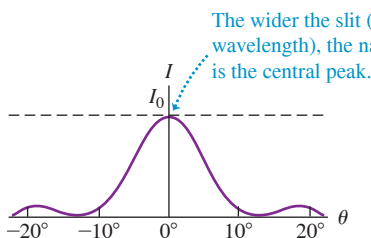
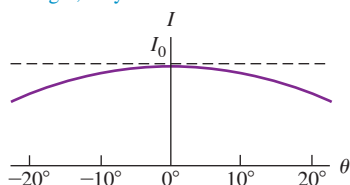
Figure 36.10 The single-slit diffraction pattern depends on the ratio of the slit width a to the wavelength λ .

(a) $a = \lambda$

(b) $a = 5\lambda$

(c) $a = 8\lambda$

If the slit width is equal to or narrower than the wavelength, only one broad maximum forms.



It's important to keep in mind that diffraction occurs for *all* kinds of waves, not just light. Sound waves undergo diffraction when they pass through a slit or aperture such as an ordinary doorway. The sound waves used in speech have wavelengths of about a meter or greater, and a typical doorway is less than 1 m wide; in this situation, a is less than λ , and the central intensity maximum extends over 180° . This is why the sounds coming through an open doorway can easily be heard by an eavesdropper hiding out of sight around the corner. In the same way, sound waves can bend around the head of an instructor who faces the blackboard while lecturing (**Fig. 36.11**). By contrast, there is essentially no diffraction of visible light through a doorway because the width a is very much greater than the wavelength λ (of order 5×10^{-7} m). You can *hear* around corners because typical sound waves have relatively long wavelengths; you cannot *see* around corners because the wavelength of visible light is very short.

Figure 36.11 The sound waves used in speech have a long wavelength (about 1 m) and can easily bend around this instructor's head. By contrast, light waves have very short wavelengths and undergo very little diffraction. Hence you can't *see* around his head!



EXAMPLE 36.2 Single-slit diffraction: Intensity I

WITH VARIATION PROBLEMS

(a) The intensity at the center of a single-slit diffraction pattern is I_0 . What is the intensity at a point in the pattern where there is a 66 radian phase difference between wavelets from the two edges of the slit? (b) If this point is 7.0° away from the central maximum, how many wavelengths wide is the slit?

IDENTIFY and SET UP In our analysis of Fig. 36.8 we used the symbol β for the phase difference between wavelets from the two edges of the slit. In part (a) we use Eq. (36.5) to find the intensity I at the point in the pattern where $\beta = 66$ rad. In part (b) we need to find the slit width a as a multiple of the wavelength λ so our target variable is a/λ . We are given the angular position θ of the point where $\beta = 66$ rad, so we can use Eq. (36.6) to solve for a/λ .

EXECUTE (a) We have $\beta/2 = 33$ rad, so from Eq. (36.5),

$$I = I_0 \left[\frac{\sin(33 \text{ rad})}{33 \text{ rad}} \right]^2 = (9.2 \times 10^{-4}) I_0$$

(b) From Eq. (36.6),

$$\frac{a}{\lambda} = \frac{\beta}{2\pi \sin \theta} = \frac{66 \text{ rad}}{(2\pi \text{ rad}) \sin 7.0^\circ} = 86$$

For example, for 550 nm light the slit width is $a = (86)(550 \text{ nm}) = 4.7 \times 10^{-5} \text{ m} = 0.047 \text{ mm}$, or roughly $\frac{1}{20} \text{ mm}$.

EVALUATE To what point in the diffraction pattern does this value of β correspond? To find out, note that $\beta = 66$ rad is approximately equal to 21π . This is an odd multiple of π , corresponding to the form $(2m + 1)\pi$ found in Eq. (36.9) for the intensity *maxima*. Hence $\beta = 66$ rad corresponds to a point near the tenth ($m = 10$) maximum. This is well beyond the range shown in Fig. 36.9a, which shows only maxima out to $m = \pm 3$.

KEYCONCEPT The intensity at a given point in a single-slit diffraction pattern depends on the phase difference β between the wavelets that arrive at that point from the two edges of the slit [Eqs. (36.6) and (36.7)]. The intensity is greatest at the center of the pattern where $\beta = 0$; the intensity is zero at points where $\beta = \pm 2\pi, \pm 4\pi, \pm 6\pi, \dots$

EXAMPLE 36.3 Single-slit diffraction: Intensity I

WITH VARIATION PROBLEMS

In the experiment described in Example 36.1 (Section 36.2), the intensity at the center of the pattern is I_0 . What is the intensity at a point on the screen 3.0 mm from the center of the pattern?

IDENTIFY and SET UP This is similar to Example 36.2, except that we are not given the value of the phase difference β at the point in question. We use geometry to determine the angle θ for our point and then use Eq. (36.7) to find the intensity I (the target variable).

EXECUTE Referring to Fig. 36.5a, we have $y = 3.0 \text{ mm}$ and $x = 6.0 \text{ m}$, so $\tan \theta = y/x = (3.0 \times 10^{-3} \text{ m})/(6.0 \text{ m}) = 5.0 \times 10^{-4}$. This is so small that the values of $\tan \theta$, $\sin \theta$, and θ (in radians) are all nearly the same. Then, from Eq. (36.7),

$$\frac{\pi a \sin \theta}{\lambda} = \frac{\pi(2.4 \times 10^{-4} \text{ m})(5.0 \times 10^{-4})}{6.33 \times 10^{-7} \text{ m}} = 0.60$$

$$I = I_0 \left(\frac{\sin 0.60}{0.60} \right)^2 = 0.89 I_0$$

EVALUATE Figure 36.9a shows that an intensity this high can occur only within the central intensity maximum. This checks out; from Example 36.1, the first intensity minimum ($m = 1$ in Fig. 36.9a) is $(32 \text{ mm})/2 = 16 \text{ mm}$ from the center of the pattern, so the point in question here at $y = 3 \text{ mm}$ does, indeed, lie within the central maximum.

KEYCONCEPT When light of wavelength λ passes through a single slit of width a , you can write the intensity of the diffracted light at angle θ [Eq. (36.7)] in terms of the ratio $(\pi a \sin \theta)/\lambda$, which is one-half of the phase difference β between wavelets coming from the two edges of the slit. The intensity is greatest at the center of the pattern ($\theta = 0$); the intensity is zero where $(\pi a \sin \theta)/\lambda = \pm \pi, \pm 2\pi, \pm 3\pi, \dots$

TEST YOUR UNDERSTANDING OF SECTION 36.3 Coherent electromagnetic radiation is sent through a slit of width 0.0100 mm . For which of the following wavelengths will there be *no* points in the diffraction pattern where the intensity is zero? (i) Blue light of wavelength 500 nm ; (ii) infrared light of wavelength $10.6\text{ }\mu\text{m}$; (iii) microwaves of wavelength 1.00 mm ; (iv) ultraviolet light of wavelength 50.0 nm .

ANSWER

(i) and (ii) If the slit width a is less than the wavelength λ , there are no points in the diffraction pattern at which the intensity is zero (see Fig. 36.10a). The slit width is 0.0100 mm . (iii) This condition is satisfied for $\lambda = 10.6\text{ }\mu\text{m}$ and $\lambda = 1.00\text{ mm}$. (iv) For $\lambda = 50.0\text{ nm}$ but not for $\lambda = 500\text{ nm}$. (v) For $\lambda = 50.0\text{ nm}$ but not for $\lambda = 500\text{ nm}$.

36.4 MULTIPLE SLITS

In Sections 35.2 and 35.3 we analyzed interference from two point sources and from two very narrow slits; in this analysis we ignored effects due to the finite (that is, nonzero) slit width. In Sections 36.2 and 36.3 we considered the diffraction effects that occur when light passes through a single slit of finite width. Additional interesting effects occur when we have two slits with finite width or when there are several very narrow slits.

Two Slits of Finite Width

Let's take another look at the two-slit pattern in the more realistic case in which the slits have finite width. If the slits are narrow in comparison to the wavelength, we can assume that light from each slit spreads out uniformly in all directions to the right of the slit. We used this assumption in Section 35.3 to calculate the interference pattern described by Eq. (35.10) or (35.15), consisting of a series of equally spaced, equally intense maxima. However, when the slits have finite width, the peaks in the two-slit interference pattern are modulated by the single-slit diffraction pattern characteristic of the width of each slit.

Figure 36.12a shows the intensity in a single-slit diffraction pattern with slit width a . The *diffraction minima* are labeled by the integer $m_d = \pm 1, \pm 2, \dots$ (“d” for “diffraction”). **Figure 36.12b** shows the pattern formed by two very narrow slits with distance d between slits, where d is four times as great as the single-slit width a in Fig. 36.12a; that is, $d = 4a$. The *interference maxima* are labeled by the integer $m_i = 0, \pm 1, \pm 2, \dots$ (“i” for “interference”). We note that the spacing between adjacent minima in the single-slit pattern is four times as great as in the two-slit pattern. Now suppose we widen each of the narrow slits to the same width a as that of the single slit in Fig. 36.12a. **Figure 36.12c** shows the pattern from two slits with width a , separated by a distance (between centers) $d = 4a$. The effect of the finite width of the slits is to superimpose the two patterns—that is, to multiply the two intensities at each point. The two-slit peaks are in the same positions as before, but their intensities are modulated by the single-slit pattern, which acts as an “envelope” for the intensity function. The expression for the intensity shown in Fig. 36.12c is proportional to the product of the two-slit and single-slit expressions, Eqs. (35.10) and (36.5):

$$I = I_0 \cos^2 \frac{\phi}{2} \left[\frac{\sin(\beta/2)}{\beta/2} \right]^2 \quad (\text{two slits of finite width}) \quad (36.12)$$

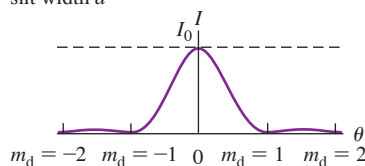
where, as before,

$$\phi = \frac{2\pi d}{\lambda} \sin \theta \quad \beta = \frac{2\pi a}{\lambda} \sin \theta$$

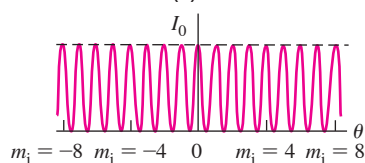
In Fig. 36.12c, every fourth interference maximum at the sides is *missing* because these interference maxima ($m_i = \pm 4, \pm 8, \dots$) coincide with diffraction minima ($m_d = \pm 1, \pm 2, \dots$). This can also be seen in Fig. 36.12d, which is a photograph of an actual pattern with $d = 4a$. You should be able to convince yourself that there will be “missing” maxima whenever d is an integer multiple of a .

Figure 36.12 Finding the intensity pattern for two slits of finite width.

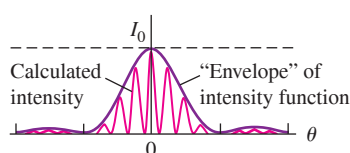
(a) Single-slit diffraction pattern for a slit width a



(b) Two-slit interference pattern for narrow slits whose separation d is four times the width of the slit in (a)



(c) Calculated intensity pattern for two slits of width a and separation $d = 4a$, including both interference and diffraction effects



(d) Photograph of the pattern calculated in (c)



For $d = 4a$, every fourth interference maximum at the sides ($m_i = \pm 4, \pm 8, \dots$) is missing.

Figures 36.12c and 36.12d show that as you move away from the central bright maximum of the two-slit pattern, the intensity of the maxima decreases. This is a result of the single-slit modulating pattern shown in Fig. 36.12a; mathematically, the decrease in intensity arises from the factor $(\beta/2)^2$ in the denominator of Eq. (36.12). You can also see this decrease in Fig. 35.6 (Section 35.2). The narrower the slits, the broader the single-slit pattern (as in Fig. 36.10) and the slower the decrease in intensity from one interference maximum to the next.

Shall we call the pattern in Fig. 36.12d *interference* or *diffraction*? It's really both, since it results from the superposition of waves coming from various parts of the two apertures.

Several Slits

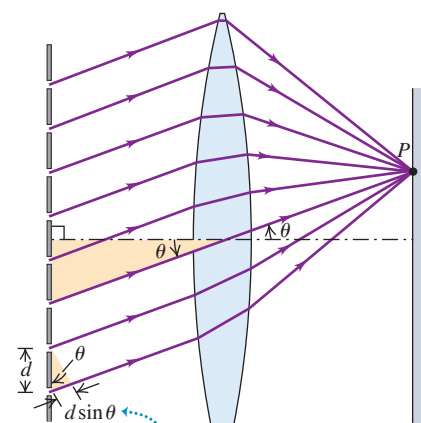
Next let's consider patterns produced by *several* very narrow slits. As we'll see, systems of narrow slits are of tremendous practical importance in *spectroscopy*, the determination of the particular wavelengths of light coming from a source. Assume that each slit is narrow in comparison to the wavelength, so its diffraction pattern spreads out nearly uniformly. **Figure 36.13** shows an array of eight narrow slits, with distance d between adjacent slits. Constructive interference occurs for rays at angle θ to the normal that arrive at point P with a path difference between adjacent slits equal to an integer number of wavelengths:

$$d \sin \theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots)$$

This means that reinforcement occurs when the phase difference ϕ at P for light from adjacent slits is an integer multiple of 2π . That is, the maxima in the pattern occur at the *same* positions as for two slits with the same spacing.

What happens *between* the maxima is different with multiple slits, however. In the two-slit pattern, there is exactly one intensity minimum located midway between each pair of maxima, corresponding to angles for which the phase difference between waves from the two sources is π , 3π , 5π , and so on. In the eight-slit pattern these are also minima because the light from adjacent slits cancels out in pairs, corresponding to the phasor diagram in **Fig. 36.14a**. But these are not the only minima in the eight-slit pattern. For example, when the phase difference ϕ from adjacent sources is $\pi/4$, the phasor diagram is as shown in Fig. 36.14b; the total (resultant) phasor is zero, and the intensity is zero. When $\phi = \pi/2$, we get the phasor diagram of Fig. 36.14c, and again both the total phasor and the intensity are zero. More generally, **the intensity with eight slits is zero whenever ϕ is an integer multiple of $\pi/4$, except when ϕ is a multiple of 2π** . Thus there are seven minima for every maximum.

Figure 36.13 Multiple-slit diffraction. Here a lens is used to give a Fraunhofer pattern on a nearby screen, as in Fig. 36.4d.



Maxima occur where the path difference for adjacent slits is a whole number of wavelengths: $d \sin \theta = m\lambda$.

Figure 36.14 Phasor diagrams for light passing through eight narrow slits. Intensity maxima occur when the phase difference $\phi = 0, 2\pi, 4\pi, \dots$. Between the maxima at $\phi = 0$ and $\phi = 2\pi$ are seven minima, corresponding to $\phi = \pi/4, \pi/2, 3\pi/4, \pi, 5\pi/4, 3\pi/2$, and $7\pi/4$. Can you draw phasor diagrams for the other minima?

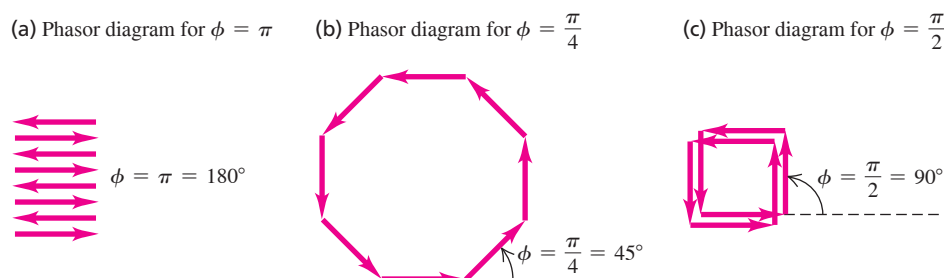
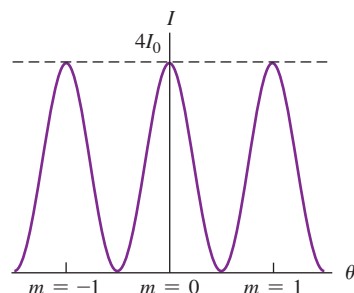
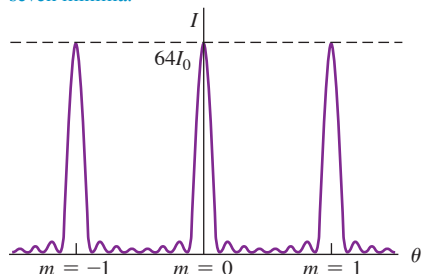


Figure 36.15 Interference patterns for N equally spaced, very narrow slits. (a) Two slits. (b) Eight slits. (c) Sixteen slits. The vertical scales are different for each graph; the maximum intensity is I_0 for a single slit and $N^2 I_0$ for N slits. The width of each peak is proportional to $1/N$.

(a) $N = 2$: two slits produce one minimum between adjacent maxima.



(b) $N = 8$: eight slits produce taller, narrower maxima in the same locations, separated by seven minima.



(c) $N = 16$: with 16 slits, the maxima are even taller and narrower, with more intervening minima.

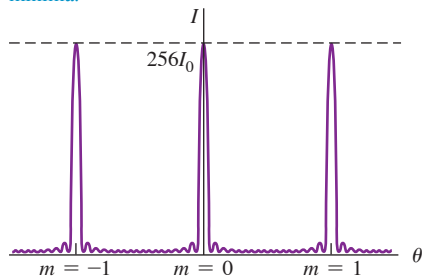


Figure 36.15b shows the result of a detailed calculation of the eight-slit pattern. The large maxima, called *principal maxima*, are in the same positions as for the two-slit pattern of Fig. 36.15a but are much narrower. If the phase difference ϕ between adjacent slits is slightly different from a multiple of 2π , the waves from slits 1 and 2 will be only a little out of phase; however, the phase difference between slits 1 and 3 will be greater, that between slits 1 and 4 will be greater still, and so on. This leads to a partial cancellation for angles that are only slightly different from the angle for a maximum, giving the narrow maxima in Fig. 36.15b. The maxima are even narrower with 16 slits (Fig. 36.15c).

You should show that when there are N slits, there are $(N - 1)$ minima between each pair of principal maxima and a minimum occurs whenever ϕ is an integer multiple of $2\pi/N$ (except when ϕ is an integer multiple of 2π , which gives a principal maximum). There are small *secondary* intensity maxima between the minima; these become smaller in comparison to the principal maxima as N increases. The greater the value of N , the narrower the principal maxima become. From an energy standpoint the total power in the entire pattern is proportional to N . The height of each principal maximum is proportional to N^2 , so from energy conservation the width of each principal maximum must be proportional to $1/N$. In the next section we'll see why the details of the multiple-slit pattern are of great practical importance.

TEST YOUR UNDERSTANDING OF SECTION 36.4 Suppose two slits, each of width a , are separated by a distance $d = 2.5a$. Are there any missing maxima in the interference pattern produced by these slits? If so, which are missing? If not, why not?

ANSWER Yes; $m = \pm 5, \pm 10, \dots$. A "missing maximum" satisfies both $d \sin \theta = m\lambda$ (the condition for an interference maximum) and $a \sin \theta = m'\lambda$ (the condition for a diffraction minimum). Substituting $d = 2.5a$, we can combine these two conditions into the relationship $m = 2.5m'$. This is satisfied for $m = \pm 5$ and $m = \pm 10$ (the fifth interference maximum is missing because it coincides with the second diffraction minimum), $m = \pm 15$ and $m = \pm 20$ (the tenth interference maximum is missing because it coincides with the fourth diffraction minimum), and so on.

36.5 THE DIFFRACTION GRATING

We have just seen that increasing the number of slits in an interference experiment (while keeping the spacing of adjacent slits constant) gives interference patterns in which the maxima are in the same positions, but progressively narrower, than with two slits. Because these maxima are so narrow, their angular position, and hence the wavelength, can be measured to very high precision. As we'll see, this effect has many important applications.

An array of a large number of parallel slits, all with the same width a and spaced equal distances d between centers, is called a **diffraction grating**. The first one was constructed by Fraunhofer using fine wires. Gratings can be made by using a diamond point to scratch many equally spaced grooves on a glass or metal surface, or by photographic reduction of a pattern of black and white stripes on paper. For a grating, what we have been calling *slits* are often called *rulings* or *lines*.

In Fig. 36.16, GG' is a cross section of a *transmission grating*; the slits are perpendicular to the plane of the page, and an interference pattern is formed by the light that is transmitted through the slits. The diagram shows only six slits; an actual grating may contain several thousand. The spacing d between centers of adjacent slits is called the *grating spacing*. A plane monochromatic wave is incident normally on the grating from the left side. We assume far-field (Fraunhofer) conditions; that is, the pattern is formed on a screen that is far enough away that all rays emerging from the grating and going to a particular point on the screen can be considered to be parallel.

We found in Section 36.4 that the principal intensity maxima with multiple slits occur in the same directions as for the two-slit pattern. These are the directions for which the path difference for adjacent slits is an integer number of wavelengths. So the positions of the maxima are once again

$$d \sin \theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \quad (36.13)$$

Intensity maxima, multiple slits:
 Distance between slits d
 Wavelength λ
 Angle of line from center of slit array to m th bright region on screen

The intensity patterns for two, eight, and 16 slits displayed in Fig. 36.15 show the progressive increase in sharpness of the maxima as the number of slits increases.

When a grating containing hundreds or thousands of slits is illuminated by a beam of parallel rays of monochromatic light, the pattern is a series of very sharp lines at angles determined by Eq. (36.13). The $m = \pm 1$ lines are called the *first-order lines*, the $m = \pm 2$ lines the *second-order lines*, and so on. **If the grating is illuminated by white light with a continuous distribution of wavelengths, each value of m corresponds to a continuous spectrum in the pattern.** The angle for each wavelength is determined by Eq. (36.13); for a given value of m , long wavelengths (the red end of the spectrum) lie at larger angles (that is, are deviated more from the straight-ahead direction) than do the shorter wavelengths at the violet end of the spectrum.

As Eq. (36.13) shows, the sines of the deviation angles of the maxima are proportional to the ratio λ/d . For substantial deviation to occur, the grating spacing d should be of the same order of magnitude as the wavelength λ . Gratings for use with visible light (λ from 400 to 700 nm) usually have about 1000 slits per millimeter; the value of d is the *reciprocal* of the number of slits per unit length, so d is of the order of $\frac{1}{1000} \text{ mm} = 1000 \text{ nm}$.

In a *reflection grating*, the array of equally spaced slits shown in Fig. 36.16 is replaced by an array of equally spaced ridges or grooves on a reflective screen. The reflected light has maximum intensity at angles where the phase difference between light waves reflected from adjacent ridges or grooves is an integer multiple of 2π . If light of wavelength λ is incident normally on a reflection grating with a spacing d between adjacent ridges or grooves, the *reflected* angles at which intensity maxima occur are given by Eq. (36.13).

The rainbow-colored reflections from the surface of a DVD are a reflection-grating effect (**Fig. 36.17**). The “grooves” are tiny pits $0.12 \mu\text{m}$ deep in the surface of the disc, with a uniform radial spacing of $0.74 \mu\text{m} = 740 \text{ nm}$. Information is coded on the DVD by varying the *length* of the pits. The reflection-grating aspect of the disc is merely an aesthetic side benefit.

Figure 36.16 A portion of a transmission diffraction grating. The separation between the centers of adjacent slits is d .

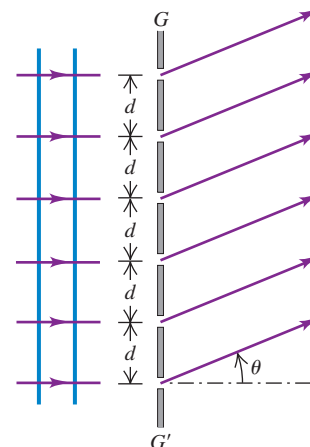


Figure 36.17 Microscopic pits on the surface of this DVD act as a reflection grating, splitting white light into its component colors.



EXAMPLE 36.4 Width of a grating spectrum

WITH VARIATION PROBLEMS

The wavelengths of the visible spectrum are approximately 380 nm (violet) to 750 nm (red). (a) Find the angular limits of the first-order visible spectrum produced by a plane grating with 600 slits per millimeter when white light falls normally on the grating. (b) Do the first-order and second-order spectra overlap? What about the second-order and third-order spectra? Do your answers depend on the grating spacing?

IDENTIFY and SET UP We must find the angles spanned by the visible spectrum in the first-, second-, and third-order spectra. These correspond to $m = 1, 2$, and 3 in Eq. (36.13).

EXECUTE (a) The grating spacing is

$$d = \frac{1}{600 \text{ slits/mm}} = 1.67 \times 10^{-6} \text{ m}$$

We solve Eq. (36.13) for θ :

$$\theta = \arcsin \frac{m\lambda}{d}$$

Then for $m = 1$, the angular deviations θ_{v1} and θ_{r1} for violet and red light, respectively, are

$$\theta_{v1} = \arcsin \left(\frac{380 \times 10^{-9} \text{ m}}{1.67 \times 10^{-6} \text{ m}} \right) = 13.2^\circ$$

$$\theta_{r1} = \arcsin \left(\frac{750 \times 10^{-9} \text{ m}}{1.67 \times 10^{-6} \text{ m}} \right) = 26.7^\circ$$

That is, the first-order visible spectrum appears with deflection angles from $\theta_{v1} = 13.2^\circ$ (violet) to $\theta_{r1} = 26.7^\circ$ (red).

Continued

(b) With $m = 2$ and $m = 3$, our equation $\theta = \arcsin(m\lambda/d)$ for 380 nm violet light yields

$$\theta_{v2} = \arcsin\left(\frac{2(380 \times 10^{-9} \text{ m})}{1.67 \times 10^{-6} \text{ m}}\right) = 27.1^\circ$$

$$\theta_{v3} = \arcsin\left(\frac{3(380 \times 10^{-9} \text{ m})}{1.67 \times 10^{-6} \text{ m}}\right) = 43.0^\circ$$

For 750 nm red light, this same equation gives

$$\theta_{r2} = \arcsin\left(\frac{2(750 \times 10^{-9} \text{ m})}{1.67 \times 10^{-6} \text{ m}}\right) = 63.9^\circ$$

$$\theta_{r3} = \arcsin\left(\frac{3(750 \times 10^{-9} \text{ m})}{1.67 \times 10^{-6} \text{ m}}\right) = \arcsin(1.35) = \text{undefined}$$

Hence the second-order spectrum extends from 27.1° to 63.9° and the third-order spectrum extends from 43.0° to 90° (the largest possible value of θ). The undefined value of θ_{r3} means that the third-order spectrum

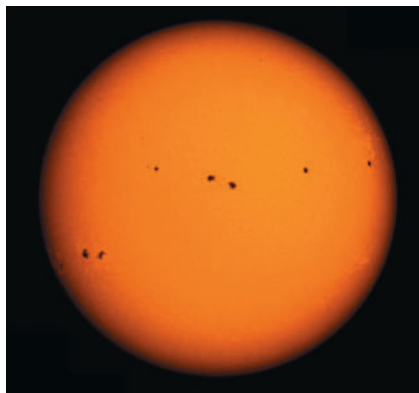
reaches $\theta = 90^\circ = \arcsin(1)$ at a wavelength shorter than 750 nm; you should be able to show that this happens for $\lambda = 557 \text{ nm}$. Hence the first-order spectrum (from 13.2° to 26.7°) does not overlap with the second-order spectrum, but the second- and third-order spectra do overlap. You can convince yourself that this is true for any value of the grating spacing d .

EVALUATE The fundamental reason the first-order and second-order visible spectra don't overlap is that the human eye is sensitive to only a narrow range of wavelengths. Can you show that if the eye could detect wavelengths from 380 nm to 900 nm (in the near-infrared range), the first and second orders *would* overlap?

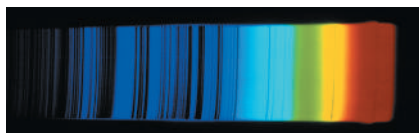
KEYCONCEPT In the interference pattern produced by a transmission or reflection grating, the intensity maxima occur at angles that are determined by the grating spacing and the wavelength of the light used. These angles are the same as in the interference pattern produced in the double-slit experiment; the difference is that the maxima are much narrower for a grating.

Figure 36.18 (a) A visible-light photograph of the sun. (b) Sunlight is dispersed into a spectrum by a diffraction grating. Specific wavelengths are absorbed as sunlight passes through the sun's atmosphere, leaving dark lines in the spectrum.

(a)



(b)



Grating Spectrographs

Diffraction gratings are widely used to measure the spectrum of light emitted by a source, a process called *spectroscopy* or *spectrometry*. Light incident on a grating of known spacing is dispersed into a spectrum. The angles of deviation of the maxima are then measured, and Eq. (36.13) is used to compute the wavelength. With a grating that has many slits, very sharp maxima are produced, and the angle of deviation (and hence the wavelength) can be measured very precisely.

An important application of this technique is to astronomy. As light generated within the sun passes through the sun's atmosphere, certain wavelengths are selectively absorbed. The result is that the spectrum of sunlight produced by a diffraction grating has dark *absorption lines* (Fig. 36.18). Experiments in the laboratory show that different types of atoms and ions absorb light of different wavelengths. By comparing these laboratory results with the wavelengths of absorption lines in the spectrum of sunlight, astronomers can deduce the chemical composition of the sun's atmosphere. The same technique is used to make chemical assays of galaxies that are millions of light-years away.

Figure 36.19 shows one design for a *grating spectrograph* used in astronomy. A transmission grating is used in the figure; in other setups, a reflection grating is used. In older designs a prism was used rather than a grating, and a spectrum was formed by dispersion (see Section 33.4) rather than diffraction. However, there is no simple relationship between wavelength and angle of deviation for a prism, prisms absorb some of the light that passes through them, and they are less effective for many nonvisible wavelengths that are important in astronomy. For these and other reasons, gratings are preferred in precision applications.

Resolution of a Grating Spectrograph

In spectroscopy it is often important to distinguish slightly differing wavelengths. The minimum wavelength difference $\Delta\lambda$ that can be distinguished by a spectrograph is described by the **chromatic resolving power** R , defined as

$$R = \frac{\lambda}{\Delta\lambda} \quad (\text{chromatic resolving power}) \quad (36.14)$$

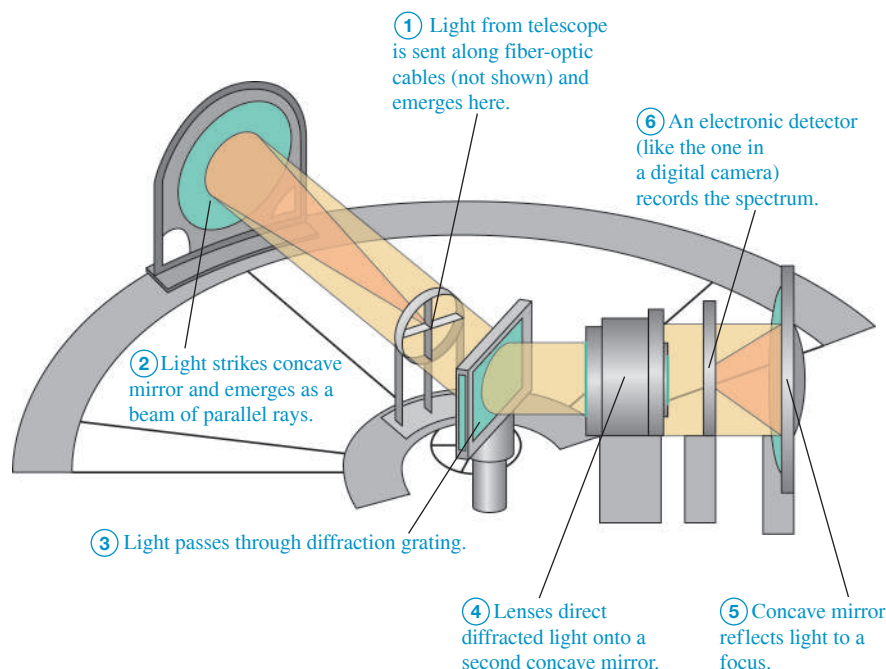


Figure 36.19 A schematic diagram of a diffraction-grating spectrograph for use in astronomy. Note that the light does not strike the grating normal to its surface, so the intensity maxima are given by a somewhat different expression than Eq. (36.13).

As an example, when sodium atoms are heated, they emit strongly at the yellow wavelengths 589.00 nm and 589.59 nm. A spectrograph that can barely distinguish these two lines in the spectrum (called the *sodium doublet*) has a chromatic resolving power $R = (589.00 \text{ nm})/(0.59 \text{ nm}) = 1000$. (You can see these wavelengths when boiling water with table salt in it on a gas range. If the water boils over onto the flame, dissolved sodium from the salt emits a burst of yellow light.)

We can derive an expression for the resolving power of a diffraction grating used in a spectrograph. Two different wavelengths give diffraction maxima at slightly different angles. As a reasonable (though arbitrary) criterion, let's assume that we can distinguish them as two separate peaks if the maximum of one coincides with the first minimum of the other.

From our discussion in Section 36.4 the m th-order maximum occurs when the phase difference ϕ for adjacent slits is $\phi = 2\pi m$. The first minimum beside that maximum occurs when $\phi = 2\pi m + 2\pi/N$, where N is the number of slits. The phase difference is also given by $\phi = (2\pi d \sin \theta)/\lambda$, so the angular interval $d\theta$ corresponding to a small increment $d\phi$ in the phase shift can be obtained from the differential of this equation:

$$d\phi = \frac{2\pi d \cos \theta d\theta}{\lambda}$$

When $d\phi = 2\pi/N$, this corresponds to the angular interval $d\theta$ between a maximum and the first adjacent minimum. Thus $d\theta$ is given by

$$\frac{2\pi}{N} = \frac{2\pi d \cos \theta d\theta}{\lambda} \quad \text{or} \quad d \cos \theta d\theta = \frac{\lambda}{N}$$

Now we need to find the angular spacing $d\theta$ between maxima for two slightly different wavelengths. The positions of these maxima are given by $d \sin \theta = m\lambda$, and the differential of this equation gives

$$d \cos \theta d\theta = m d\lambda$$

According to our criterion, the limit or resolution is reached when these two angular spacings are equal. Equating the two expressions for the quantity $(d \cos \theta d\theta)$, we find

$$\frac{\lambda}{N} = m d\lambda \quad \text{and} \quad \frac{\lambda}{d\lambda} = Nm$$

BIO APPLICATION Detecting DNA with Diffraction

Diffraction gratings are used in a common piece of laboratory equipment known as a spectrophotometer. Light shining across a diffraction grating is dispersed into its component wavelengths. A slit is used to block all but a very narrow range of wavelengths, producing a beam of almost perfectly monochromatic light. The instrument then measures how much of that light is absorbed by a solution of biological molecules. For example, the sample tube shown here contains a solution of DNA, which is transparent to visible light but which strongly absorbs ultraviolet light with a wavelength of exactly 260 nm. Therefore, by illuminating the sample with 260 nm light and measuring the amount absorbed, we can determine the concentration of DNA in the solution.



CAUTION Watch out for different uses of the symbol d Don't confuse the slit spacing d with the differential " d " in the angular interval $d\theta$ or in the phase shift increment $d\phi$!

If $\Delta\lambda$ is small, we can replace $d\lambda$ by $\Delta\lambda$, and the resolving power R is

$$R = \frac{\lambda}{\Delta\lambda} = Nm \quad (36.15)$$

The greater the number of slits N , the better the resolution; also, the higher the order m of the diffraction-pattern maximum that we use, the better the resolution.

TEST YOUR UNDERSTANDING OF SECTION 36.5 What minimum number of slits would be required in a grating to resolve the sodium doublet in the fourth order? (i) 250; (ii) 400; (iii) 1000; (iv) 4000.

ANSWER

(i) As described in the text, the resolving power needed is $R = Nm = 1000$. In the first order ($m = 1$) we need $N = 1000$ slits, but in the fourth order ($m = 4$) we need only $N = 1000/4 = 250$ slits. (These numbers are only approximate because of the arbitrary nature of our criterion for resolution and because real gratings always have slight imperfections in the shapes and spacings of the slits.)

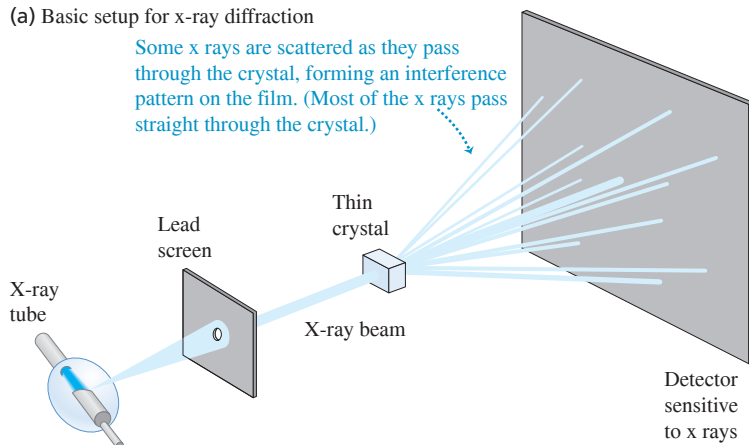
36.6 X-RAY DIFFRACTION

X rays were discovered by Wilhelm Röntgen (1845–1923) in 1895, and early experiments suggested that they were electromagnetic waves with wavelengths of the order of 10^{-10} m. At about the same time, the idea began to emerge that in a crystalline solid the atoms are arranged in a regular repeating pattern, with spacing between adjacent atoms also of the order of 10^{-10} m. Putting these two ideas together, Max von Laue (1879–1960) proposed in 1912 that a crystal might serve as a kind of three-dimensional diffraction grating for x rays. That is, a beam of x rays might be scattered (that is, absorbed and re-emitted) by the individual atoms in a crystal, and the scattered waves might interfere just like waves from a diffraction grating.

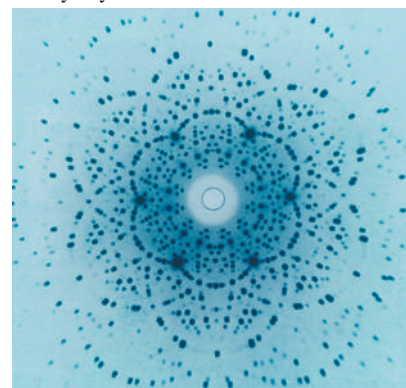
The first **x-ray diffraction** experiments were performed in 1912 by Friedrich, Knipping, and von Laue, using the experimental setup shown in **Fig. 36.20a**. The scattered x rays *did* form an interference pattern, which they recorded on photographic film. Figure 36.20b is a photograph of such a pattern. These experiments verified that x rays *are* waves, or at least have wavelike properties, and also that the atoms in a crystal *are* arranged in a regular pattern (**Fig. 36.21**). Since that time, x-ray diffraction has proved to be an invaluable research tool, both for measuring x-ray wavelengths and for studying the structure of crystals and complex molecules.

Figure 36.20 (a) An x-ray diffraction experiment. (b) Diffraction pattern (or *Laue pattern*) formed by directing a beam of x rays at a thin section of beryl crystal.

(a) Basic setup for x-ray diffraction



(b) Laue diffraction pattern for a thin section of beryl crystal



A Simple Model of X-Ray Diffraction

To better understand x-ray diffraction, we consider first a two-dimensional scattering situation, as shown in Fig. 36.22a, in which a plane wave is incident on a rectangular array of scattering centers. The situation might be a ripple tank with an array of small posts or x rays incident on an array of atoms. In the case of electromagnetic waves, the wave induces an oscillating electric dipole moment in each scatterer. These dipoles act like little antennas, emitting scattered waves. The resulting interference pattern is the superposition of all these scattered waves. The situation is different from that with a diffraction grating, in which the waves from all the slits are emitted *in phase* (for a plane wave at normal incidence). Here the scattered waves are *not* all in phase because their distances from the *source* are different. To compute the interference pattern, we have to consider the *total* path differences for the scattered waves, including the distances from source to scatterer and from scatterer to observer.

As Fig. 36.22b shows, the path length from source to observer is the same for all the scatterers in a single row if the two angles θ_a and θ_r are equal. Scattered radiation from *adjacent* rows is *also* in phase if the path difference for adjacent rows is an integer number of wavelengths. Figure 36.22c shows that this path difference is $2d\sin\theta$, where θ is the common value of θ_a and θ_r . Therefore the conditions for radiation from the *entire* array to reach the observer in phase are (1) the angle of incidence must equal the angle of scattering and (2) the path difference for adjacent rows must equal $m\lambda$, where m is an integer. We can express the second condition, called the **Bragg condition** in honor of x-ray diffraction pioneers Sir William Bragg and his son Laurence Bragg, as

Bragg condition for constructive interference from an array:

$$2d\sin\theta = m\lambda \quad (m = 1, 2, 3, \dots) \quad (36.16)$$

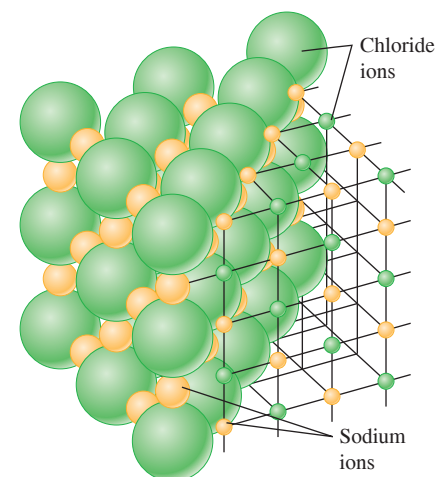
Distance between adjacent rows in array

Wavelength

Angle of line from surface of array to m th bright region on screen

In directions for which Eq. (36.16) is satisfied, we see a strong maximum in the interference pattern. We can describe this interference in terms of *reflections* of the wave from the horizontal rows of scatterers in Fig. 36.22a. Strong reflection (constructive interference) occurs at angles such that the incident and scattered angles are equal and Eq. (36.16) is satisfied. Since $\sin\theta$ can never be greater than 1, Eq. (36.16) says that to have constructive interference the quantity $m\lambda$ must be less than $2d$ and so λ must be less than $2d/m$. For example, the value of d in an NaCl crystal (see Fig. 36.21) is only 0.282 nm. Hence to have the m th-order maximum present in the diffraction pattern, λ must be less than $2(0.282 \text{ nm})/m$; that is, $\lambda < 0.564 \text{ nm}$ for $m = 1$, $\lambda < 0.282 \text{ nm}$ for $m = 2$, $\lambda < 0.188 \text{ nm}$ for $m = 3$, and so on. These are all x-ray wavelengths (see Fig. 32.4), which is why x rays are used for studying crystal structure.

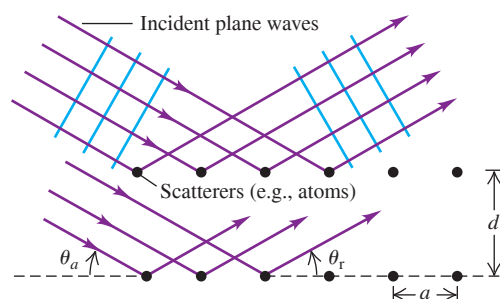
Figure 36.21 Model of the arrangement of ions in a crystal of NaCl (table salt). The spacing of adjacent atoms is 0.282 nm. (The electron clouds of the atoms actually overlap slightly.)



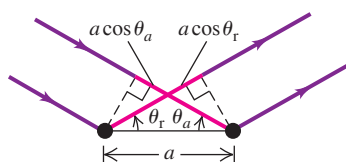
CAUTION Scattering from an array In Eq. (36.16) the angle θ is measured with respect to the *surface* of the crystal rather than with respect to the *normal* to the plane of an array of slits or a grating. Also, note that the path difference in Eq. (36.16) is $2d\sin\theta$, not $d\sin\theta$ as in Eq. (36.13) for a diffraction grating. ■

Figure 36.22 A two-dimensional model of scattering from a rectangular array. The distance between adjacent atoms in a horizontal row is a ; the distance between adjacent rows is d . The angles in (b) are measured from the *surface* of the array, not from its normal.

(a) Scattering of waves from a rectangular array



(b) Scattering from adjacent atoms in a row is constructive when the path lengths $a\cos\theta_a$ and $a\cos\theta_r$ are equal, so that the angle of incidence θ_a equals the angle of reflection (scattering) θ_r .



(c) Scattering from atoms in adjacent rows is constructive when the path difference $2d\sin\theta$ is an integer number of wavelengths, as in Eq. (36.16).

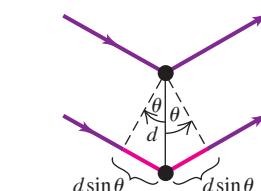
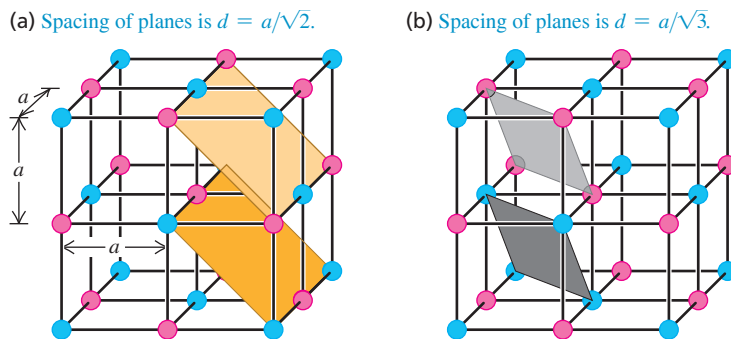
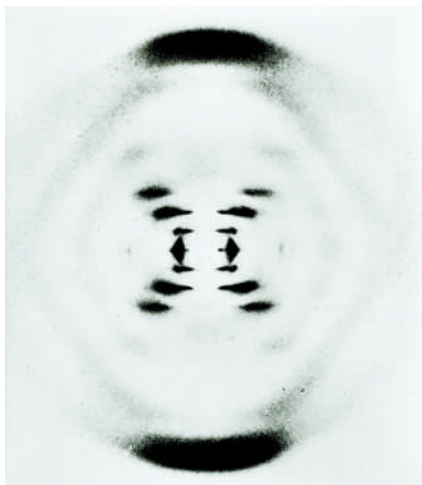


Figure 36.23 A cubic crystal and two different families of crystal planes. There are also three sets of planes parallel to the cube faces, with spacing a .

CAUTION Bragg reflection is really Bragg interference. While we are using the term *reflection*, remember that we are dealing with an *interference* effect. The reflections from various planes are closely analogous to interference effects in thin films (see Section 35.4).

Figure 36.24 The British scientist Rosalind Franklin made this groundbreaking x-ray diffraction image of DNA in 1953. The dark bands arranged in a cross provided the first evidence of the helical structure of the DNA molecule.



We can extend this discussion to a three-dimensional array by considering *planes* of scatterers instead of *rows*. Figure 36.23 shows two different sets of parallel planes that pass through all the scatterers. Waves from all the scatterers in a given plane interfere constructively if the angles of incidence and scattering are equal. There is also constructive interference between planes when Eq. (36.16) is satisfied, where d is now the distance between adjacent planes. Because there are many different sets of parallel planes, there are also many values of d and many sets of angles that give constructive interference for the whole crystal lattice. This phenomenon is called **Bragg reflection**.

As Fig. 36.20b shows, in x-ray diffraction there is nearly complete cancellation in all but certain very specific directions in which constructive interference occurs and forms bright spots. Such a pattern is usually called an *x-ray diffraction pattern*, although *interference pattern* might be more appropriate.

We can determine the wavelength of x rays by examining the diffraction pattern for a crystal of known structure and known spacing between atoms, just as we determined wavelengths of visible light by measuring patterns from slits or gratings. (The spacing between atoms in simple crystals of known structure, such as sodium chloride, can be found from the density of the crystal and Avogadro's number.) Then, once we know the x-ray wavelength, we can use x-ray diffraction to explore the structure and determine the spacing between atoms in crystals with unknown structure.

X-ray diffraction is by far the most important experimental tool in the investigation of crystal structure of solids. X-ray diffraction also plays an important role in studies of the structures of liquids and of organic molecules. It has been one of the chief experimental techniques in working out the double-helix structure of DNA (Fig. 36.24) and subsequent advances in molecular genetics.

EXAMPLE 36.5 X-ray diffraction

WITH VARIATION PROBLEMS

You direct a beam of 0.154 nm x rays at certain planes of a silicon crystal. As you increase the angle of incidence of the beam from zero, the first strong interference maximum occurs when the beam makes an angle of 34.5° with the planes. (a) How far apart are the planes? (b) Will you find other interference maxima from these planes at larger angles of incidence?

IDENTIFY and SET UP This problem involves Bragg reflection of x rays from the planes of a crystal. In part (a) we use the Bragg condition, Eq. (36.16), to find the distance d between adjacent planes from the known wavelength $\lambda = 0.154$ nm and angle of incidence $\theta = 34.5^\circ$ for the $m = 1$ interference maximum. Given the value of d , we use the Bragg condition again in part (b) to find the values of θ for interference maxima corresponding to other values of m .

EXECUTE (a) We solve Eq. (36.16) for d and set $m = 1$:

$$d = \frac{m\lambda}{2 \sin \theta} = \frac{(1)(0.154 \text{ nm})}{2 \sin 34.5^\circ} = 0.136 \text{ nm}$$

This is the distance between adjacent planes.

(b) To calculate other angles, we solve Eq. (36.16) for $\sin \theta$:

$$\sin \theta = \frac{m\lambda}{2d} = m \frac{0.154 \text{ nm}}{2(0.136 \text{ nm})} = m(0.566)$$

Values of m of 2 or greater give values of $\sin \theta$ greater than unity, which is impossible. Hence there are *no* other angles for interference maxima for this particular set of crystal planes.

EVALUATE Our result in part (b) shows that there *would* be a second interference maximum if the quantity $2\lambda/2d = \lambda/d$ were less than 1. This would be the case if the wavelength of the x rays were less than $d = 0.136$ nm. How short would the wavelength need to be to have *three* interference maxima?

KEYCONCEPT In x-ray diffraction, waves that pass through or reflect from a crystal are scattered by rows of atoms within the crystal. The pattern of scattered light has maxima at points where waves that interact with different rows arrive in phase and interfere constructively [Eq. (36.16)].

TEST YOUR UNDERSTANDING OF SECTION 36.6 You are doing an x-ray diffraction experiment with a crystal in which the atomic planes are 0.200 nm apart. You are using x rays of wavelength 0.0900 nm. What is the highest-order maximum present in the diffraction pattern? (i) Third; (ii) fourth; (iii) fifth; (iv) sixth; (v) seventh.

ANSWER

(ii) The angular position of the m th maximum is given by Eq. (36.16), $2d \sin \theta = m\lambda$. This gives $m = (2d \sin \theta)/\lambda$. The sine function can never be greater than 1, so the largest value of m in the pattern can be no greater than $2d/\lambda = 2(0.200 \text{ nm})/(0.0900 \text{ nm}) = 4.44$. Since m must be an integer, the highest-order maximum in the pattern is $m = 4$ (fourth order). The $m = 5, 6, 7, \dots$ maxima do not appear.

36.7 CIRCULAR APERTURES AND RESOLVING POWER

We have studied in detail the diffraction patterns formed by long, thin slits or arrays of slits. But an aperture of *any* shape forms a diffraction pattern. The diffraction pattern formed by a *circular* aperture is of special interest because of its role in limiting how well an optical instrument can resolve fine details. In principle, we could compute the intensity at any point P in the diffraction pattern by dividing the area of the aperture into small elements, finding the resulting wave amplitude and phase at P , and then integrating over the aperture area to find the resultant amplitude and intensity at P . In practice, the integration cannot be carried out in terms of elementary functions. We'll simply *describe* the pattern and quote a few relevant numbers.

The diffraction pattern formed by a circular aperture consists of a central bright spot surrounded by a series of bright and dark rings, as **Fig. 36.25** shows. We can describe the pattern in terms of the angle θ , representing the angular radius of each ring. The angular radius θ_1 of the first *dark* ring is given by

$$\text{Diffraction by a circular aperture:} \quad \sin \theta_1 = 1.22 \frac{\lambda}{D} \quad \text{Angular radius of first dark ring} = \text{angular radius of Airy disk} \quad (36.17)$$

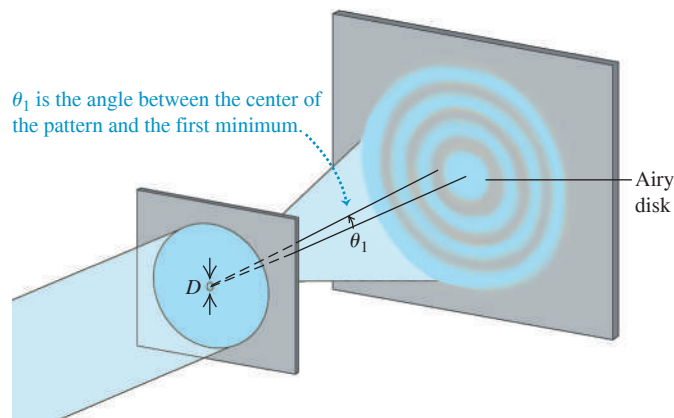
λ ← Wavelength
 D ← Aperture diameter

The angular radii of the next two dark rings are given by

$$\sin \theta_2 = 2.23 \frac{\lambda}{D} \quad \sin \theta_3 = 3.24 \frac{\lambda}{D} \quad (36.18)$$

The central bright spot is called the **Airy disk**, in honor of Sir George Airy (1801–1892), who first derived the expression for the intensity in the pattern. The angular radius of the Airy disk is that of the first dark ring, given by Eq. (36.17). The angular radii of the first three *bright* rings outside the Airy disk are

$$\sin \theta = 1.63 \frac{\lambda}{D}, \quad 2.68 \frac{\lambda}{D}, \quad 3.70 \frac{\lambda}{D} \quad (36.19)$$



APPLICATION Bigger Telescope, Better Resolving Power The large aperture diameter of very large telescopes minimizes diffraction effects. The effective diameter of a telescope can be increased by using arrays of smaller telescopes. The Very Large Array (VLA) in New Mexico is a collection of 27 radio telescopes, each 25 m in diameter, that can be spread out in a Y-shaped arrangement 36 km across. Hence the effective aperture diameter is 36 km, giving the VLA a limit of resolution of 5×10^{-8} rad at a radio wavelength of 1.5 cm. If your eye had this resolving power, you could read the “20/20” line on an eye chart more than 30 km away!



Figure 36.25 Diffraction pattern formed by a circular aperture of diameter D . The pattern consists of a central bright spot and alternating dark and bright rings. The angular radius θ_1 of the first dark ring is shown. (This diagram is not drawn to scale.)

Figure 36.26 Photograph of the diffraction pattern formed by a circular aperture.

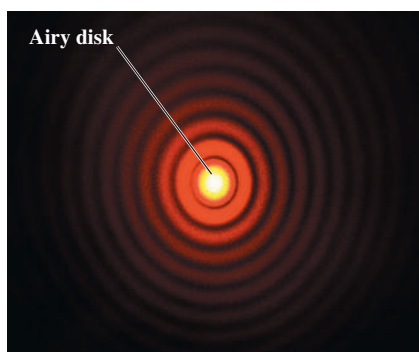
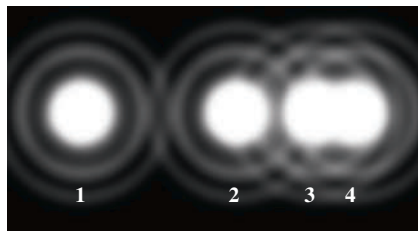
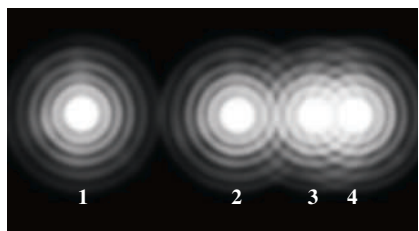


Figure 36.27 These images show the diffraction patterns produced when four very small (“point”) sources of light are viewed through a lens with a circular aperture placed in front of the lens. The photographs were made with a circular aperture in front of the lens. (a) The aperture is so small that the patterns of sources 3 and 4 overlap and are barely resolved by Rayleigh’s criterion. Increasing the size of the aperture decreases the size of the diffraction patterns, as shown in (b) and (c).

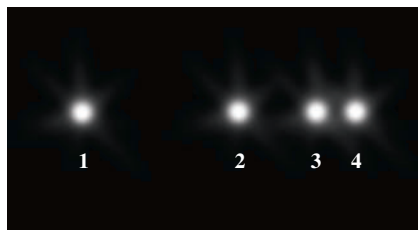
(a) Small aperture



(b) Medium aperture



(c) Large aperture



The intensities in the bright rings drop off very quickly with increasing angle. When D is much larger than the wavelength λ , the usual case for optical instruments, the peak intensity in the first ring is only 1.7% of the value at the center of the Airy disk, and the peak intensity of the second ring is only 0.4%. Most (85%) of the light energy falls within the Airy disk. **Figure 36.26** shows a diffraction pattern from a circular aperture.

Diffraction and Image Formation

Diffraction has far-reaching implications for image formation by lenses and mirrors. In our study of optical instruments in Chapter 34 we assumed that a lens with focal length f focuses a parallel beam (plane wave) to a *point* at a distance f from the lens. We now see that what we get is *not* a point but the diffraction pattern just described. If we have two point objects, their images are not two points but two diffraction patterns. When the objects are close together, their diffraction patterns overlap; if they are close enough, their patterns overlap almost completely and cannot be distinguished. The effect is shown in **Fig. 36.27**, which presents the patterns for four very small “point” sources of light. In Fig. 36.27a the image of source 1 is well separated from the others, but the images of sources 3 and 4 have merged. In Fig. 36.27b, with a larger aperture diameter and hence smaller Airy disks, the images of sources 3 and 4 are better resolved. In Fig. 36.27c, with a still larger aperture, they are well resolved.

A widely used criterion for resolution of two point objects, proposed by the English physicist Lord Rayleigh (1842–1919) and called **Rayleigh’s criterion**, is that the objects are just barely resolved (that is, distinguishable) if the center of one diffraction pattern coincides with the first minimum of the other. In that case the angular separation of the image centers is given by Eq. (36.17). The angular separation of the *objects* is the same as that of the *images* made by a telescope, microscope, or other optical device. So two point objects are barely resolved when their angular separation is given by Eq. (36.17).

The minimum separation of two objects that can just be resolved by an optical instrument is called the **limit of resolution** of the instrument. The smaller the limit of resolution, the greater the **resolution**, or **resolving power**, of the instrument. Diffraction sets the ultimate limits on resolution of lenses. *Geometric* optics may make it seem that we can make images as large as we like. Eventually, though, we always reach a point at which the image becomes larger but does not gain in detail. The images in Fig. 36.27 would not become sharper if enlarged.

CAUTION Resolving power vs. chromatic resolving power Don’t confuse the resolving power of an optical instrument with the *chromatic* resolving power of a grating (Section 36.5). Resolving power refers to the ability to distinguish the images of objects that appear close to each other, when looking either through an optical instrument or at a photograph made with the instrument. Chromatic resolving power describes how well different wavelengths can be distinguished in a spectrum formed by a diffraction grating. **I**

Rayleigh’s criterion combined with Eq. (36.17) shows that resolution (resolving power) improves with larger diameter; it also improves with shorter wavelengths. **?** Ultraviolet microscopes have higher resolution than visible-light microscopes. In electron microscopes the resolution is limited by the wavelengths associated with the electrons, which have wavelike aspects (to be discussed further in Chapter 39). These wavelengths can be made 100,000 times smaller than wavelengths of visible light, with a corresponding gain in resolution. Resolving power also explains the difference in storage capacity between DVDs (introduced in 1995) and Blu-ray discs (introduced in 2003). Information is stored in both of these in a series of tiny pits. In order not to lose information in the scanning process, the scanning optics must be able to resolve two adjacent pits so that they do not seem to blend into a single pit (see sources 3 and 4 in Fig. 36.27). The blue scanning laser used in a Blu-ray player has a shorter wavelength (405 nm) and hence better resolving power than the 650 nm red laser in a DVD player. Hence pits can be spaced closer together in a Blu-ray disc than in a DVD, and more information can be stored on a disc of the same size (50 gigabytes on a Blu-ray disc versus 4.7 gigabytes on a DVD).

BIO APPLICATION The Airy Disk in an Eagle's Eye Diffraction by the pupil of an eye limits resolving power. In a human's eye, the maximum pupil diameter D is about 5 mm; in an eagle's eye, D is about 9 mm. From Eq. (36.17) this means that an eagle's eye has superior resolution: A distant point source of light produces an Airy disk on an eagle's retina that is only about $\frac{5}{9}$ the angular size of the disk produced on the retina of the human eye. (If our eye produces an image like Fig. 36.27b, an eagle's eye produces one like Fig. 36.27c.) To record the fine details of this high-resolution image, the light-sensitive cones in an eagle's retina are smaller and more closely packed than those in a human retina.



EXAMPLE 36.6 Resolving power of a camera lens

WITH VARIATION PROBLEMS

A camera lens with focal length $f = 50$ mm and maximum aperture $f/2$ forms an image of an object 9.0 m away. (a) If the resolution is limited by diffraction, what is the minimum distance between two points on the object that are barely resolved? What is the corresponding distance between image points? (b) How does the situation change if the lens is “stopped down” to $f/16$? Use $\lambda = 500$ nm in both cases.

IDENTIFY and SET UP This example uses the ideas about resolving power, image formation by a lens (Section 34.4), and f -number (Section 34.5). From Eq. (34.20), the f -number of a lens is its focal length f divided by the aperture diameter D . We use this equation to determine D and then use Eq. (36.17) (the Rayleigh criterion) to find the angular separation θ between two barely resolved points on the object. We then use the geometry of image formation by a lens to determine the distance y between those points and the distance y' between the corresponding image points.

EXECUTE (a) The aperture diameter is $D = f/(f\text{-number}) = (50 \text{ mm})/2 = 25 \text{ mm} = 25 \times 10^{-3} \text{ m}$. From Eq. (36.17) the angular separation θ of two object points that are barely resolved is

$$\theta \approx \sin \theta = 1.22 \frac{\lambda}{D} = 1.22 \frac{500 \times 10^{-9} \text{ m}}{25 \times 10^{-3} \text{ m}} = 2.4 \times 10^{-5} \text{ rad}$$

We know from our thin-lens analysis in Section 34.4 that, apart from sign, $y/s = y'/s'$ [see Eq. (34.14)]. Thus the angular separations of the object points and the corresponding image points are both equal to θ . Because the object distance s is much greater than the focal length $f = 50$ mm, the image distance s' is approximately equal to f . Thus

$$\begin{aligned} \frac{y}{9.0 \text{ m}} &= 2.4 \times 10^{-5} & y &= 2.2 \times 10^{-4} \text{ m} = 0.22 \text{ mm} \\ \frac{y'}{50 \text{ mm}} &= 2.4 \times 10^{-5} & y' &= 1.2 \times 10^{-3} \text{ mm} \\ & & &= 0.0012 \text{ mm} \approx \frac{1}{800} \text{ mm} \end{aligned}$$

(b) The aperture diameter is now $(50 \text{ mm})/16$, or one-eighth as large as before. The angular separation between barely resolved points is eight times as great, and the values of y and y' are also eight times as great as before:

$$y = 1.8 \text{ mm} \quad y' = 0.0096 \text{ mm} = \frac{1}{100} \text{ mm}$$

Only the best camera lenses can approach this resolving power.

EVALUATE Many photographers use the smallest possible aperture for maximum sharpness, since lens aberrations cause light rays that are far from the optic axis to converge to a different image point than do rays near the axis. But as this example shows, diffraction effects become more significant at small apertures. One cause of fuzzy images has to be balanced against another.

KEYCONCEPT Like waves that pass through a narrow slit, waves that pass through a circular aperture undergo diffraction and spread out. This sets a limit on the ability of a lens or telescope to resolve objects that are close to each other.

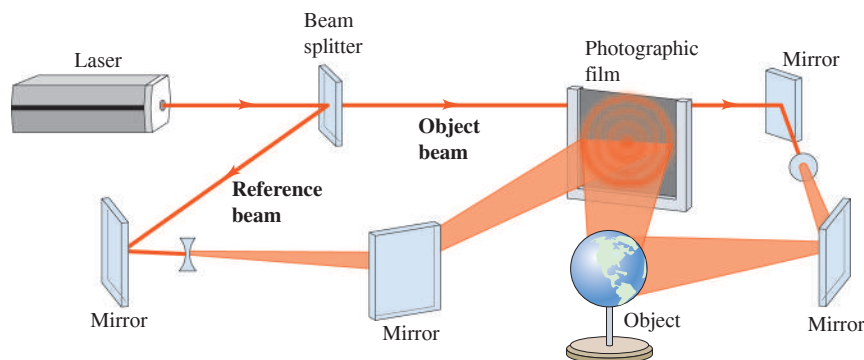
TEST YOUR UNDERSTANDING OF SECTION 36.7 You have been asked to compare four proposals for telescopes to be placed in orbit above the blurring effects of the earth's atmosphere. Rank the proposed telescopes in order of their ability to resolve small details, from best to worst. (i) A radio telescope 100 m in diameter observing at a wavelength of 21 cm; (ii) an optical telescope 2.0 m in diameter observing at a wavelength of 500 nm; (iii) an ultraviolet telescope 1.0 m in diameter observing at a wavelength of 100 nm; (iv) an infrared telescope 2.0 m in diameter observing at a wavelength of $10 \mu\text{m}$.

ANSWER

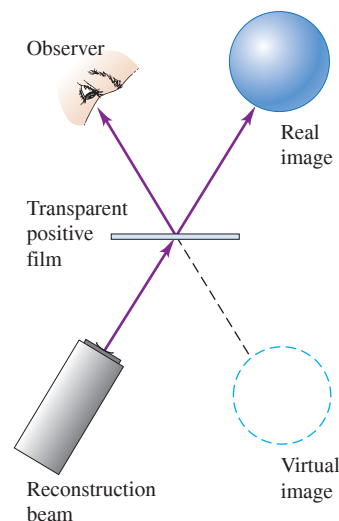
Rayleigh's criterion combined with Eq. (36.17) shows that the smaller the value of the ratio λ/D , the better the resolving power of a telescope of diameter D . For the four telescopes, this ratio is equal to (i) $(21 \text{ cm})/(100 \text{ m}) = (0.21 \text{ m})/(100 \text{ m}) = 2.1 \times 10^{-3}$; (ii) $(500 \text{ nm})/(2.0 \text{ m}) = (5.0 \times 10^{-7} \text{ m})/(2.0 \text{ m}) = 2.5 \times 10^{-7}$; (iii) $(100 \text{ nm})/(1.0 \text{ m}) = (1.0 \times 10^{-7} \text{ m})/(1.0 \text{ m}) = 1.0 \times 10^{-7}$; (iv) $(10 \mu\text{m})/(2.0 \text{ m}) = (1.0 \times 10^{-5} \text{ m})/(2.0 \text{ m}) = 5.0 \times 10^{-6}$. Ranking these ratios from smallest to largest gives (iii), (ii), (iv), (i). **|**

Figure 36.28 (a) A hologram is the record on film of the interference pattern formed with light from the coherent source and light scattered from the object. (b) Images are formed when light is projected through the hologram. The observer sees the virtual image formed behind the hologram.

(a) Recording a hologram



(b) Viewing the hologram



36.8 HOLOGRAPHY

Holography is a technique for recording and reproducing an image of an object through the use of interference effects. Unlike the two-dimensional images recorded by an ordinary photograph or television system, a holographic image is truly three-dimensional. Such an image can be viewed from different directions to reveal different sides and from various distances to reveal changing perspective. If you had never seen a hologram, you wouldn't believe it was possible!

Figure 36.28a shows the basic procedure for making a hologram. We illuminate the object to be holographed with monochromatic light, and we place a photographic film so that it is struck by scattered light from the object and also by direct light from the source. In practice, the light source must be a laser, for reasons we'll discuss later. Interference between the direct and scattered light forms a complex interference pattern that is recorded on the film.

To form the images, we simply project light through the developed film (Fig. 36.28b). Two images are formed: a virtual image on the side of the film nearer the source and a real image on the opposite side.

Holography and Interference Patterns

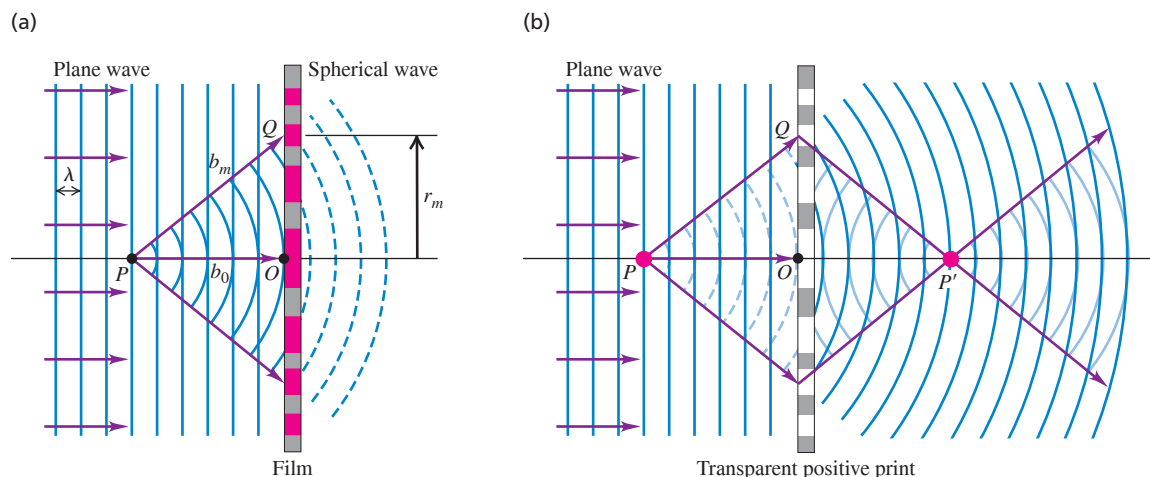
A complete analysis of holography is beyond our scope, but we can gain some insight into the process by looking at how a single point is holographed and imaged. Consider the interference pattern that is formed on a sheet of photographic negative film by the superposition of an incident plane wave and a spherical wave, as shown in Fig. 36.29a. The spherical wave originates at a point source P at a distance b_0 from the film; P may in fact be a small object that scatters part of the incident plane wave. We assume that the two waves are monochromatic and coherent and that the phase relationship is such that constructive interference occurs at point O on the diagram. Then constructive interference will *also* occur at any point Q on the film that is farther from P than O is by an integer number of wavelengths. That is, if $b_m - b_0 = m\lambda$, where m is an integer, then constructive interference occurs. The points where this condition is satisfied form circles on the film centered at O , with radii r_m given by

$$b_m - b_0 = \sqrt{b_0^2 + r_m^2} - b_0 = m\lambda \quad (m = 1, 2, 3, \dots) \quad (36.20)$$

Solving this for r_m^2 , we find

$$r_m^2 = \lambda(2mb_0 + m^2\lambda)$$

Figure 36.29 (a) Constructive interference of the plane and spherical waves occurs in the plane of the film at every point Q for which the distance b_m from P is greater than the distance b_0 from P to O by an integer number of wavelengths $m\lambda$. For the point Q shown, $m = 2$. (b) When a plane wave strikes a transparent positive print of the developed film, the diffracted wave consists of a wave converging to P' and then diverging again and a diverging wave that appears to originate at P . These waves form the real and virtual images, respectively.



Ordinarily, b_0 is very much larger than λ , so we ignore the second term in parentheses and obtain

$$r_m = \sqrt{2m\lambda b_0} \quad (m = 1, 2, 3, \dots) \quad (36.21)$$

The interference pattern consists of a series of concentric bright circular fringes with radii given by Eq. (36.21). Between these bright fringes are dark fringes.

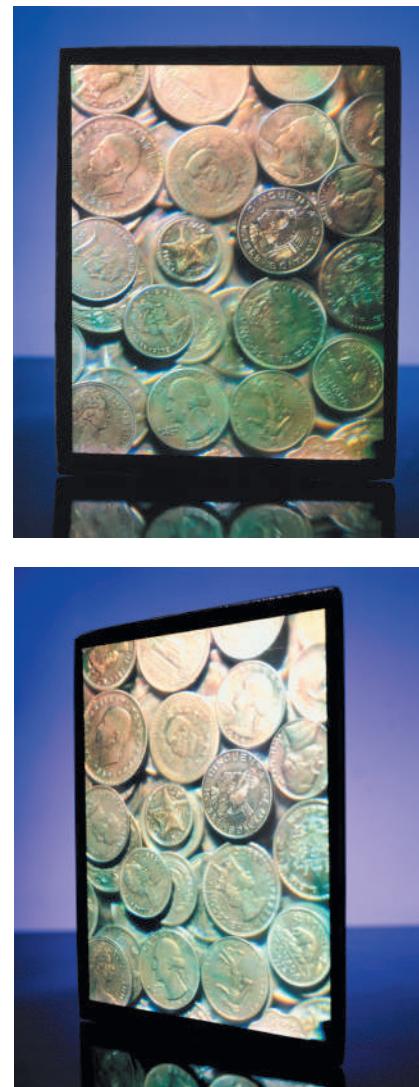
Now we develop the film and make a transparent positive print, so the bright-fringe areas have the greatest transparency on the film. Then we illuminate it with monochromatic plane-wave light of the same wavelength λ that we used initially. In Fig. 36.29b, consider a point P' at a distance b_0 along the axis from the film. The centers of successive bright fringes differ in their distances from P' by an integer number of wavelengths, and therefore a strong *maximum* in the diffracted wave occurs at P' . That is, light converges to P' and then diverges from it on the opposite side. Therefore P' is a *real image* of point P .

This is not the entire diffracted wave, however. The interference of the wavelets that spread out from all the transparent areas forms a second spherical wave that is diverging rather than converging. When this wave is traced back behind the film in Fig. 36.29b, it appears to be spreading out from point P . Thus the total diffracted wave from the hologram is a superposition of a spherical wave converging to form a real image at P' and a spherical wave that diverges as though it had come from the virtual image point P .

Because of the principle of superposition for waves, what is true for the imaging of a single point is also true for the imaging of any number of points. The film records the superposed interference pattern from the various points, and when light is projected through the film, the various image points are reproduced simultaneously. Thus the images of an extended object can be recorded and reproduced just as for a single point object. **Figure 36.30** shows photographs of a holographic image from two different angles, showing the changing perspective in this three-dimensional image.

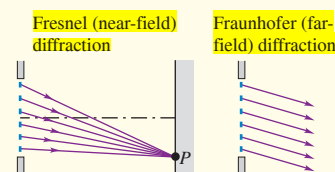
In making a hologram, we have to overcome two practical problems. First, the light used must be *coherent* over distances that are large in comparison to the dimensions of the object and its distance from the film. Ordinary light sources *do not* satisfy this requirement, for reasons that we discussed in Section 35.1. Therefore laser light is essential for making a hologram. (Ordinary white light can be used for *viewing* certain types of hologram, such as those used on credit cards.) Second, extreme mechanical stability is needed. If any relative motion of source, object, or film occurs during exposure, even by as much as a quarter of a wavelength, the interference pattern on the film is blurred enough to prevent satisfactory image formation. These obstacles are not insurmountable, however, and holography has become important in research, entertainment, and a wide variety of technological applications.

Figure 36.30 Two views of the same hologram seen from different angles.



CHAPTER 36 SUMMARY

Fresnel and Fraunhofer diffraction: Diffraction occurs when light passes through an aperture or around an edge. When the source and the observer are so far away from the obstructing surface that the outgoing rays can be considered parallel, it is called Fraunhofer diffraction. When the source or the observer is relatively close to the obstructing surface, it is Fresnel diffraction.

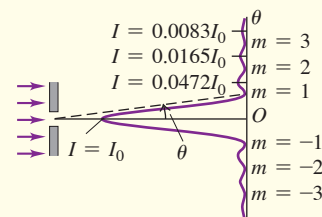


Single-slit diffraction: Monochromatic light sent through a narrow slit of width a produces a diffraction pattern on a distant screen. Equation (36.2) gives the condition for destructive interference (a dark fringe) at a point P in the pattern at angle θ . Equation (36.7) gives the intensity in the pattern as a function of θ . (See Examples 36.1–36.3.)

$$\sin \theta = \frac{m\lambda}{a} \quad (36.2)$$

$$(m = \pm 1, \pm 2, \pm 3, \dots)$$

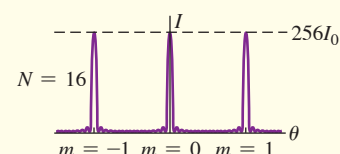
$$I = I_0 \left\{ \frac{\sin[\pi a(\sin \theta)/\lambda]}{\pi a(\sin \theta)/\lambda} \right\}^2 \quad (36.7)$$



Diffraction gratings: A diffraction grating consists of a large number of thin parallel slits, spaced a distance d apart. The condition for maximum intensity in the interference pattern is the same as for the two-source pattern, but the maxima for the grating are very sharp and narrow. (See Example 36.4.)

$$d \sin \theta = m\lambda \quad (36.13)$$

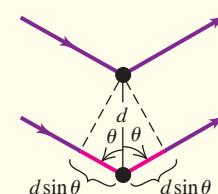
$$(m = 0, \pm 1, \pm 2, \pm 3, \dots)$$



X-ray diffraction: A crystal serves as a three-dimensional diffraction grating for x rays with wavelengths of the same order of magnitude as the spacing between atoms in the crystal. For a set of crystal planes spaced a distance d apart, constructive interference occurs when the angles of incidence and scattering (measured from the crystal planes) are equal and when the Bragg condition [Eq. (36.16)] is satisfied. (See Example 36.5.)

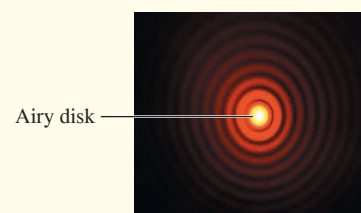
$$2d \sin \theta = m\lambda \quad (36.16)$$

$$(m = 1, 2, 3, \dots)$$



Circular apertures and resolving power: The diffraction pattern from a circular aperture of diameter D consists of a central bright spot, called the **Airy disk**, and a series of concentric dark and bright rings. Equation (36.17) gives the angular radius θ_1 of the first dark ring, equal to the angular size of the Airy disk. **Diffraction sets the ultimate limit on resolution (image sharpness) of optical instruments.** According to Rayleigh's criterion, two point objects are just barely resolved when their angular separation θ is given by Eq. (36.17). (See Example 36.6.)

$$\sin \theta_1 = 1.22 \frac{\lambda}{D} \quad (36.17)$$



Chapter 36 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLE 36.1** (Section 36.2) and **EXAMPLES 36.2 and 36.3** (Section 36.3) before attempting these problems.

VP36.3.1 You pass laser light of wavelength 645 nm through a slit 0.250 mm in width and observe the diffraction pattern on a screen a large distance away. On the screen, the centers of the second minima on either side of the central bright fringe are 28.0 mm apart. (a) How far away is the screen? (b) What would the distance be between these minima if the wavelength were 525 nm?

VP36.3.2 You shine a laser on a narrow slit 0.221 mm in width. In the diffraction pattern that appears on a screen 5.00 m from the slit, the third minimum is 45.7 mm from the middle of the central bright fringe. Find (a) the wavelength of the laser light and (b) the angle of a line from the center of the slit to the second dark fringe on the screen.

VP36.3.3 At a certain point in a single-slit diffraction pattern there is a phase difference of 35.0 radians between wavelets arriving at the point from the two edges of the slit. The slit is 0.250 mm wide, and the light used has wavelength 545 nm. (a) What is the angle of a line from the center of the slit to this point in the diffraction pattern? (b) If the intensity at the center of the diffraction pattern is I_0 , what is the intensity at this point?